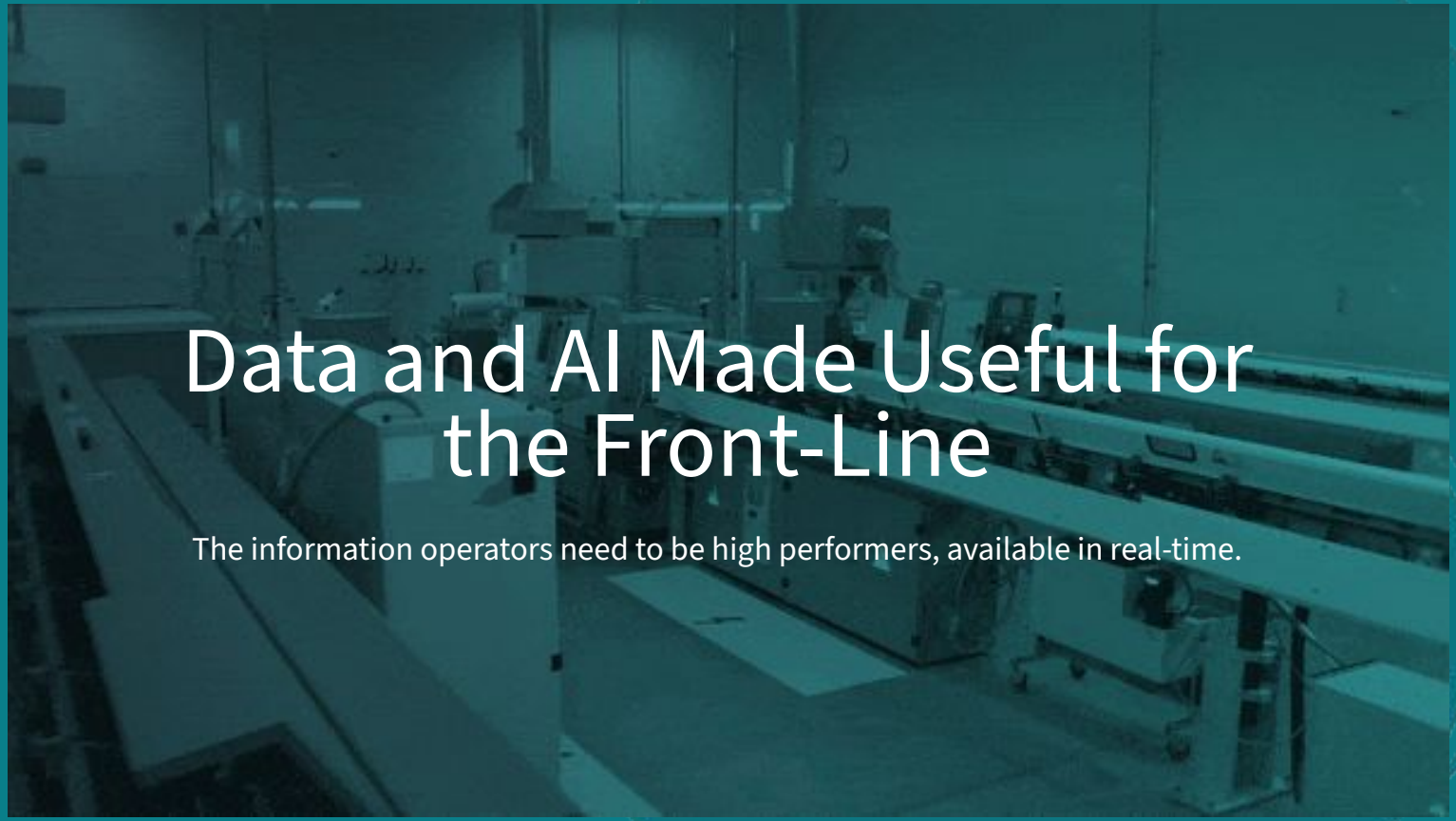


Data and AI Made Useful for the Front-Line

The information operators need to be high performers, available in real-time.

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Welcome to DevFest!

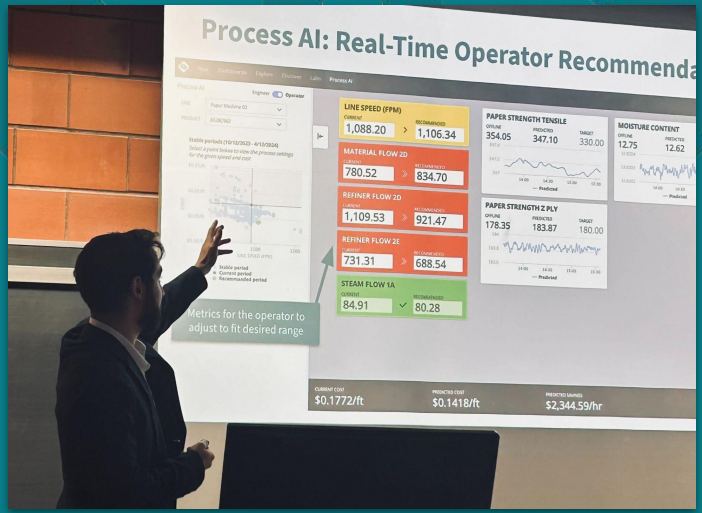


Joe Youssouf

Lead Customer Data Scientist
Oden Technologies

joseppy.ca

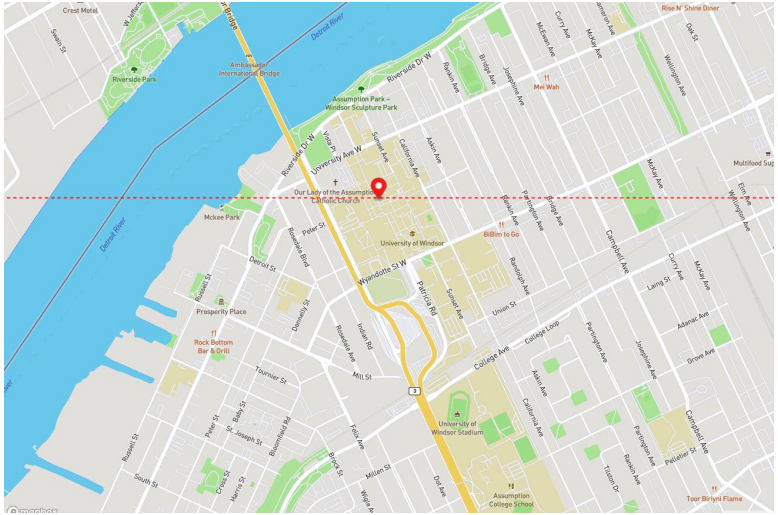




https://github.com/JYoussouf/slides/gdg-devfest-2025

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Part time Vibe Coder



How North Are You Compared to the True North Strong and Free?

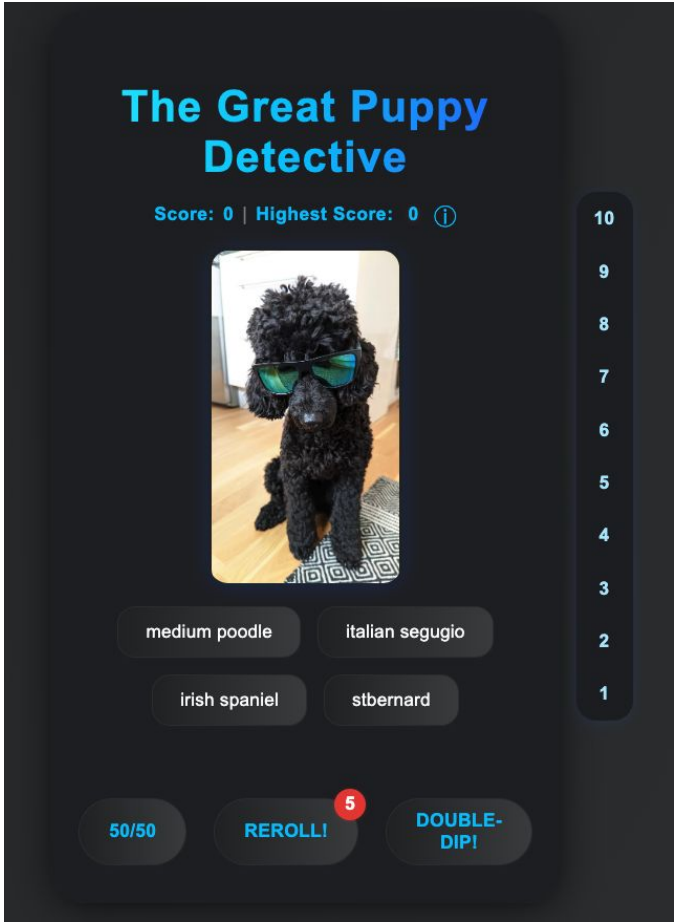
Enter your city or street to find out how many Canadians live further south than you!

This tool uses Mapbox and Canadian census data to compare your latitude to the population of Canada.

401 Sunset Avenue, Windsor, Ontario N9B 3P4, Ca

Check

You are further north than about 146,827 other Canadians, or about 0.3 percent of the country.



joseppy.ca

(big https://neal.fun fan)

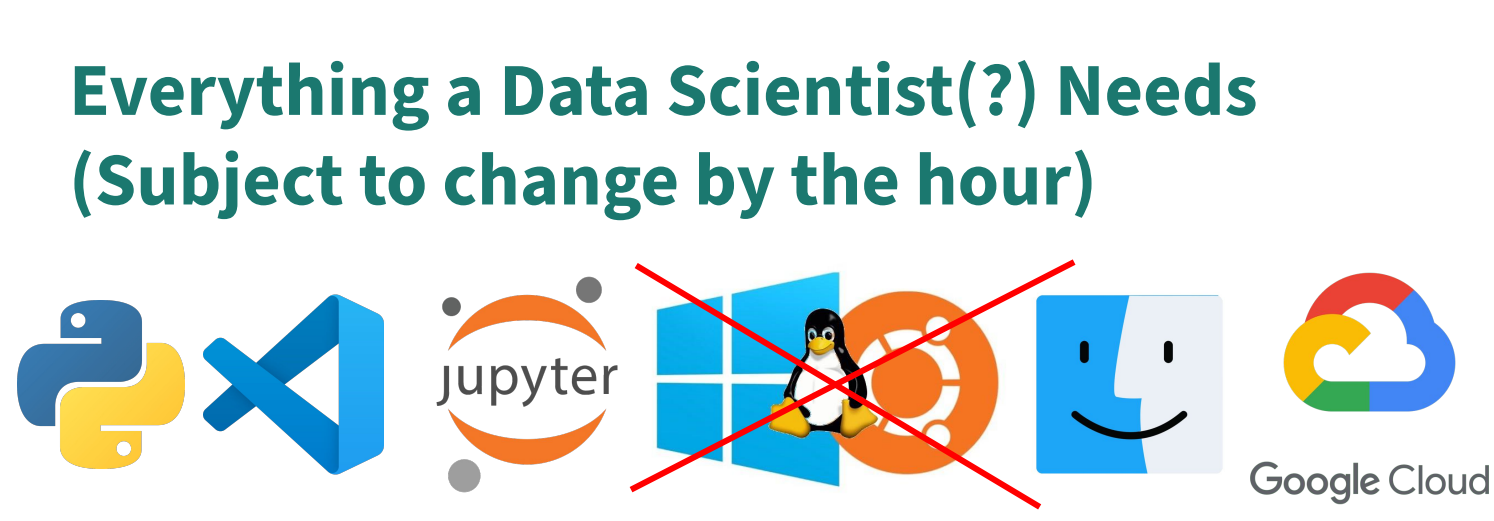
Ready Your Battle Station!







Everything a Data Scientist(?) Needs (Subject to change by the hour)



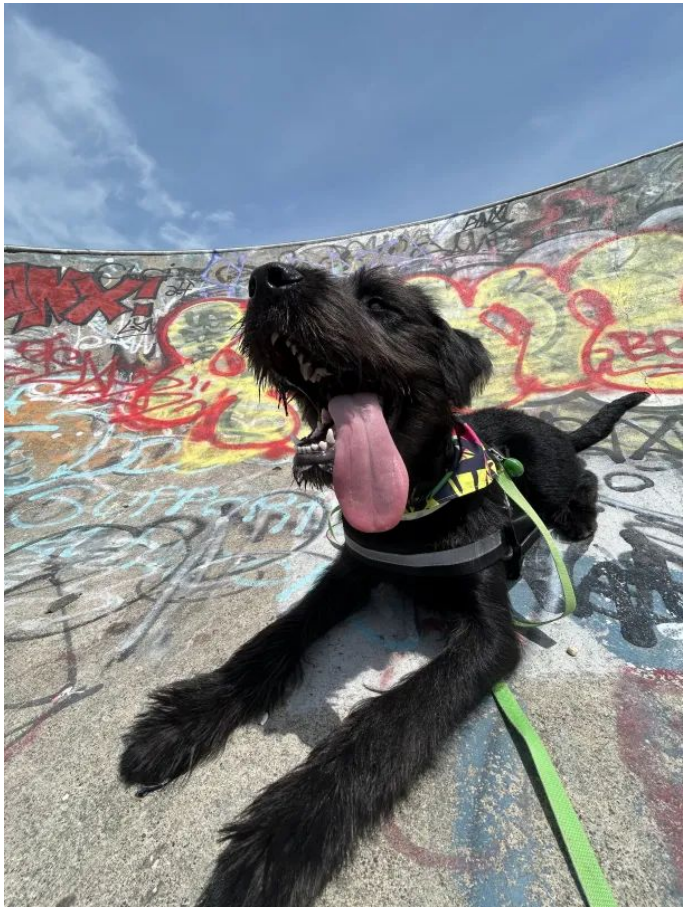


“Oden said I can play with you!”

ATLASSIAN

The best rubber duck in the world!

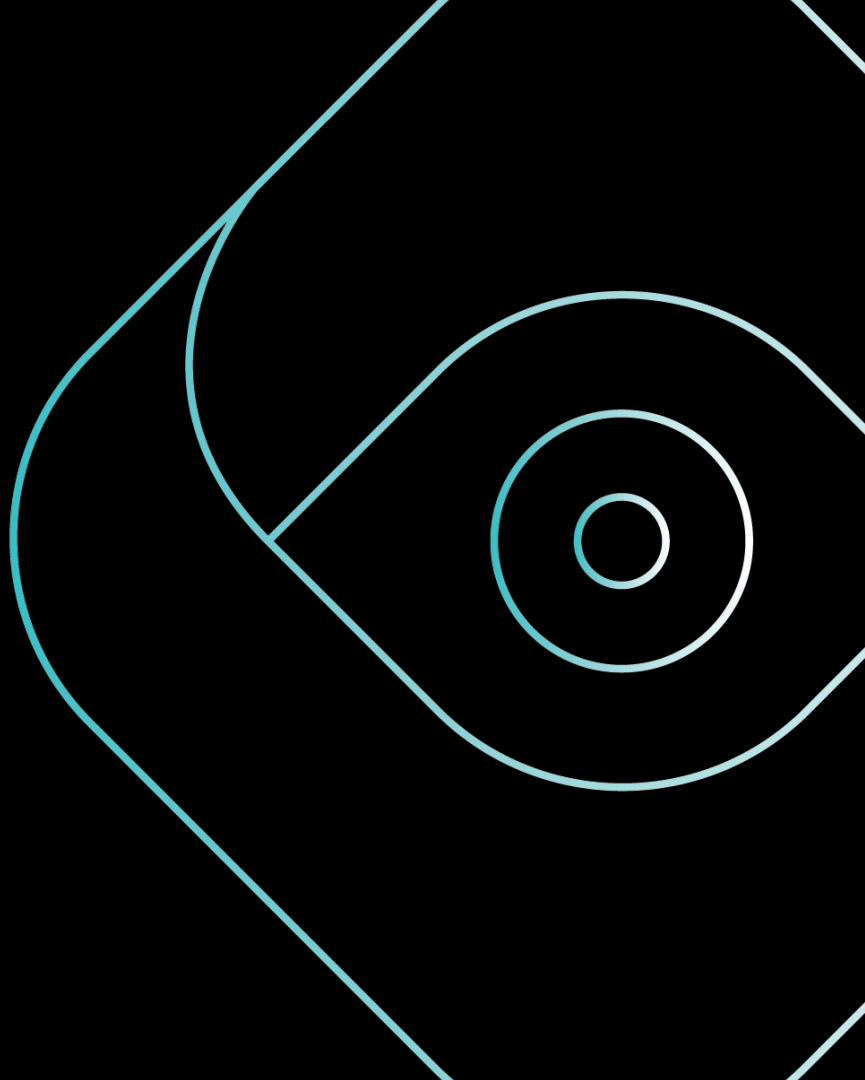




ODEN TECHNOLOGIES

Oden Technologies

Chapter 1: AI in Manufacturing?



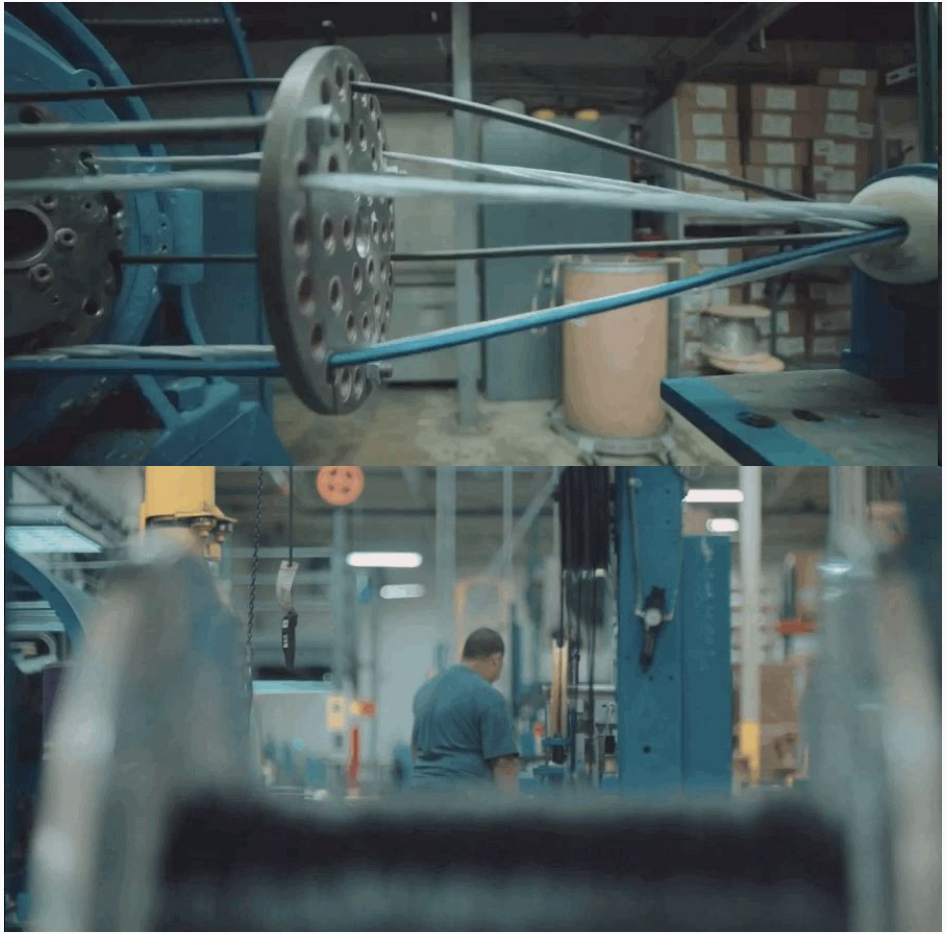
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Oden's Customers

Medium to large manufacturers in plastics extrusion, injection molding, pipes, chemical, paper and pulp.

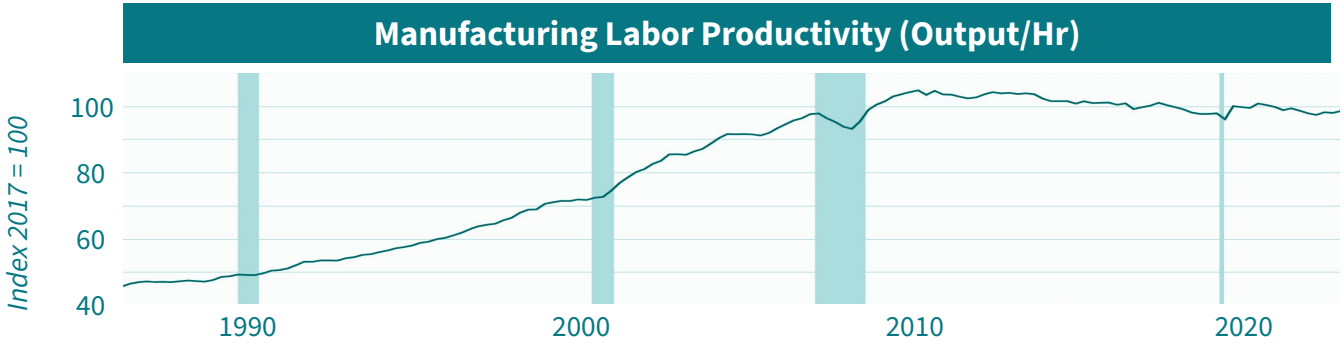
Process and Quality Engineers looking to centralize, analyze, and act on their data.

Plant managers who are looking to optimize logistics, output, and cost.



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A Perfect Storm for Manufacturers



A Typical Manufacturer's Workforce		
	2018	2025
Average Tenure	20 Years	3 Years
Average Time in Position	7 Years	9 Months
Average 3-Month Retention Rate	90%	50%

- ✖ Productivity Plateauing
- ✖ Workforce Challenges
- ✖ Demand & Asset Growth
- ✖ Margin & Competitive Pressures

Sources: LNS Research, FRED, U.S. Bureau of Labor Statistics

Front-Line Challenges

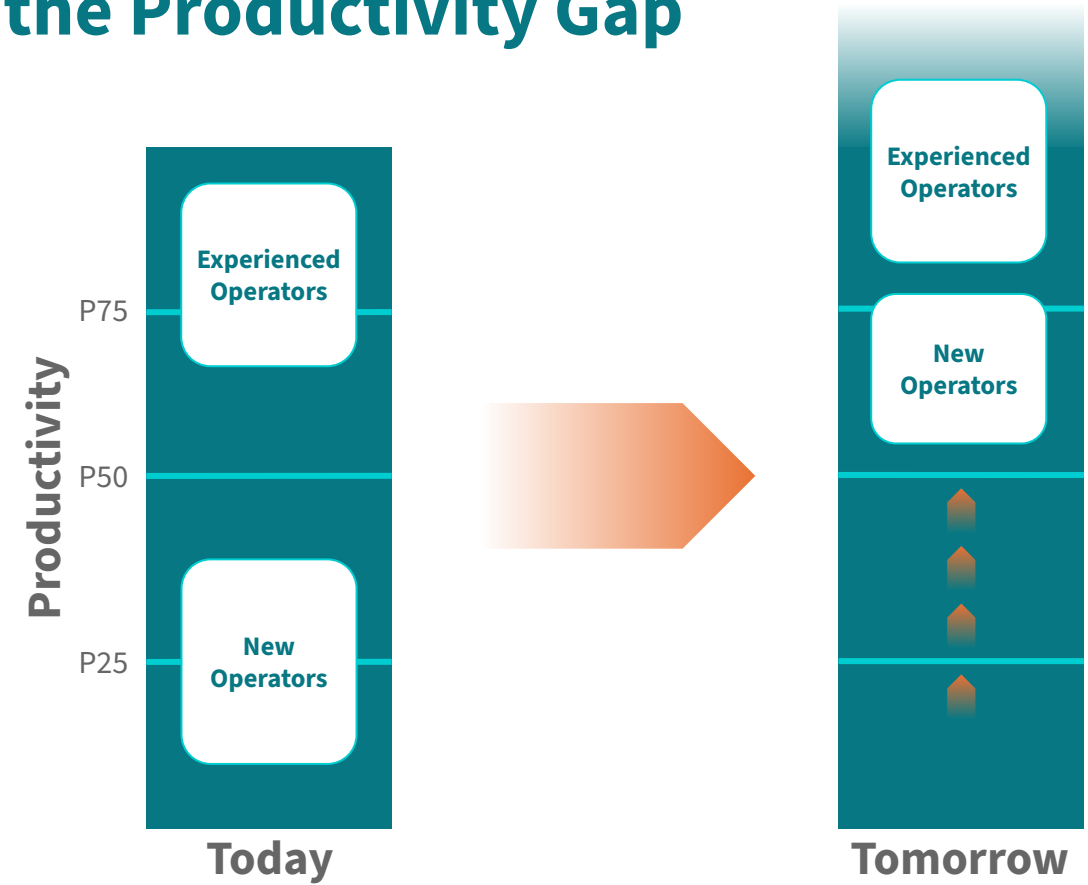
Dependent on Operator Experience

Decreasing Tenure

Losing Tribal Knowledge

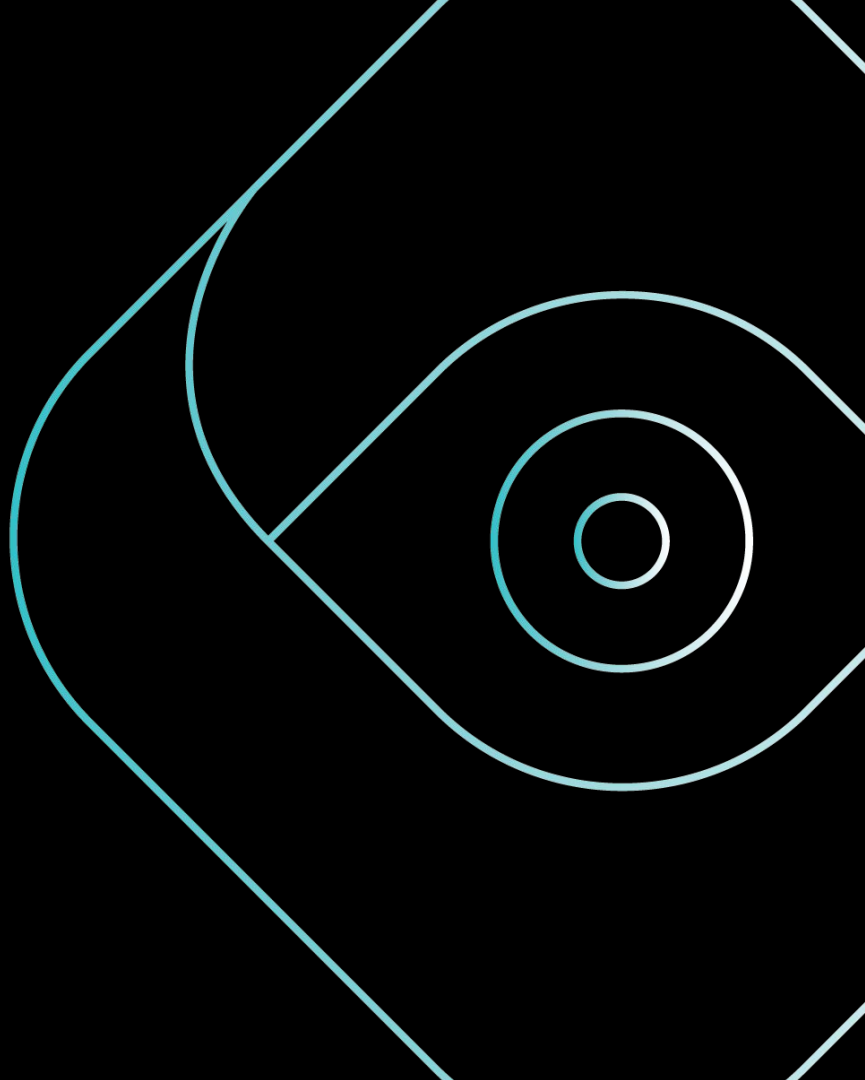


Bridging the Productivity Gap



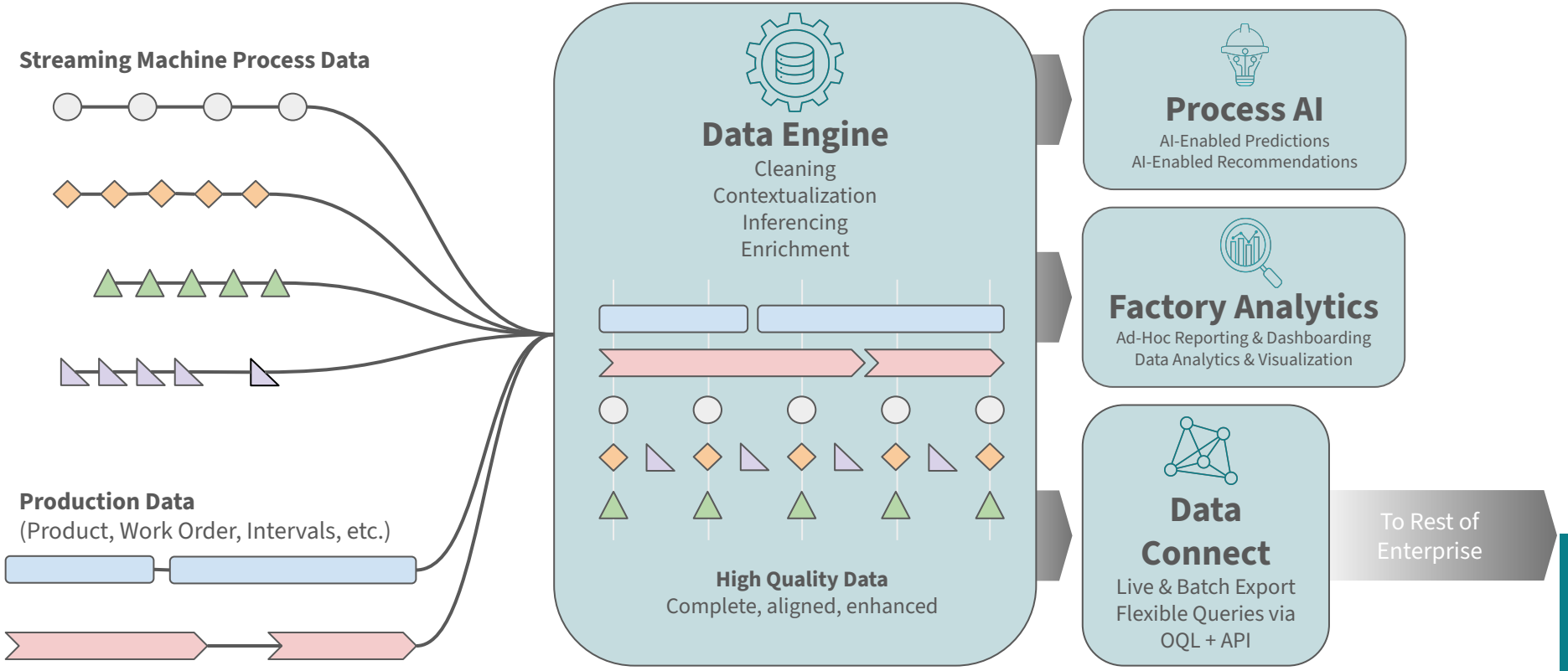
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Chapter 2: Show me the data!

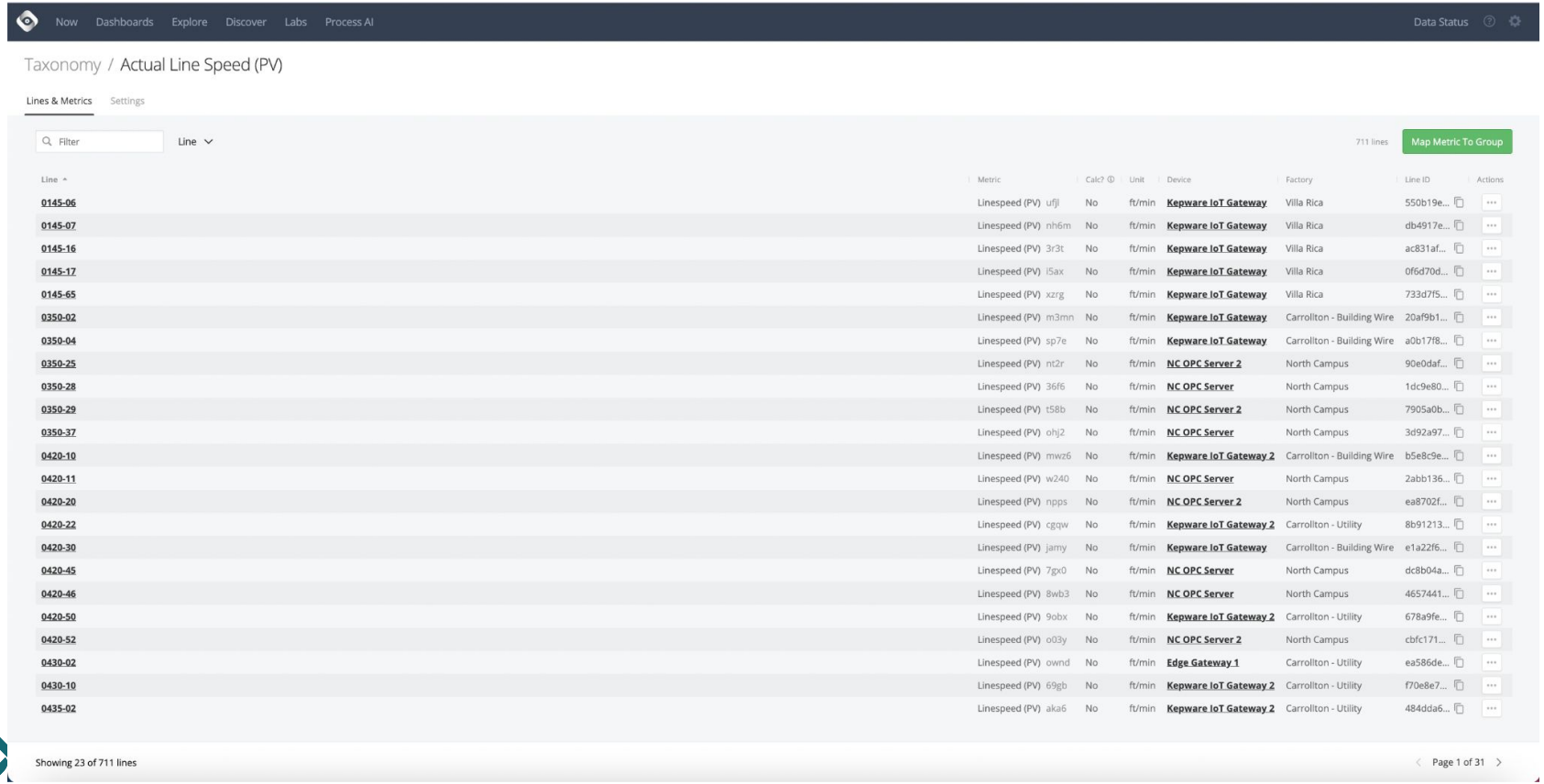


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Data Engine: Create Trusted, Contextualized Data

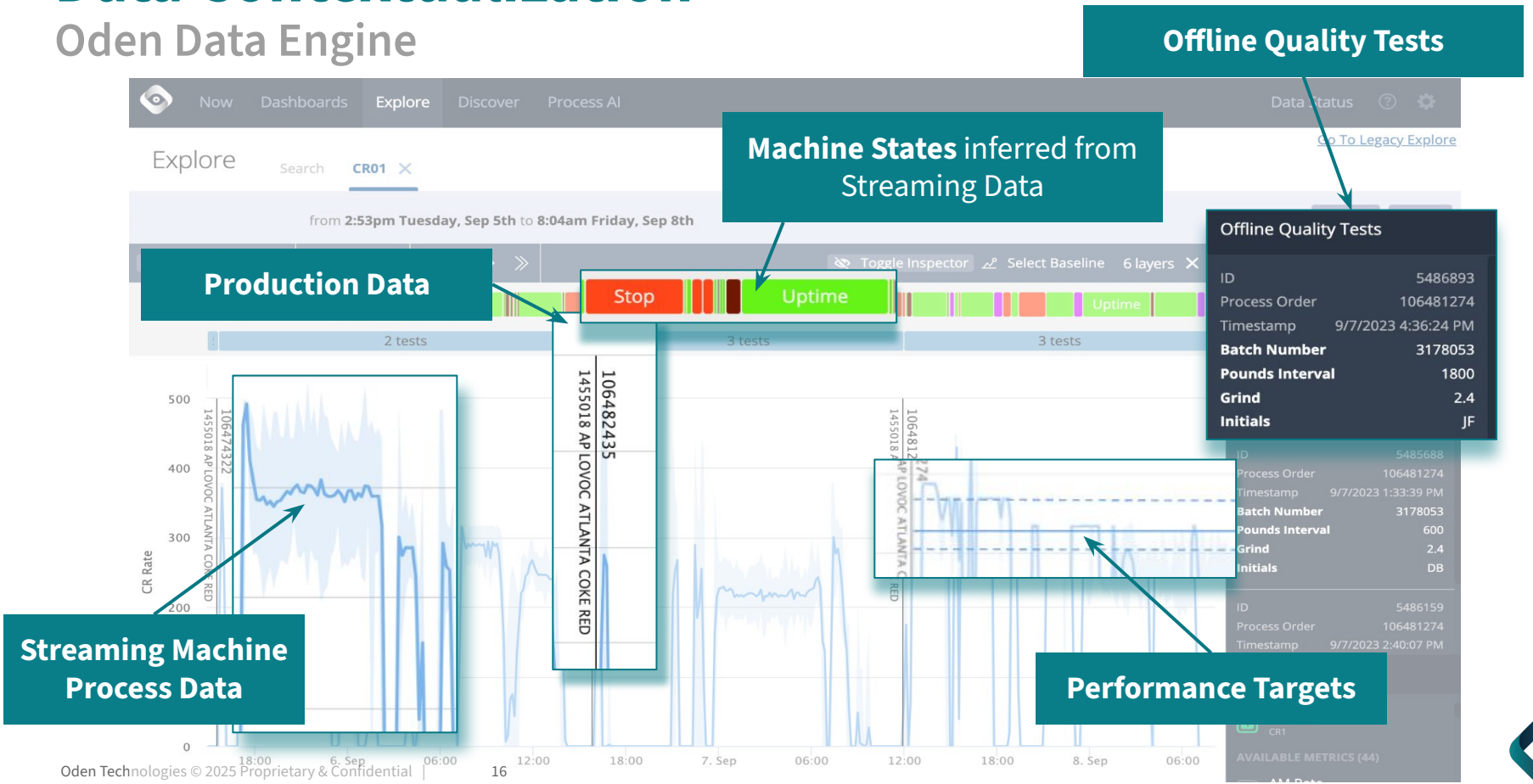


Example: Query 750+ Lines w/ Standard Naming Conventions



Data Contextualization

Oden Data Engine



35789

live-streaming time series metrics

(and counting)

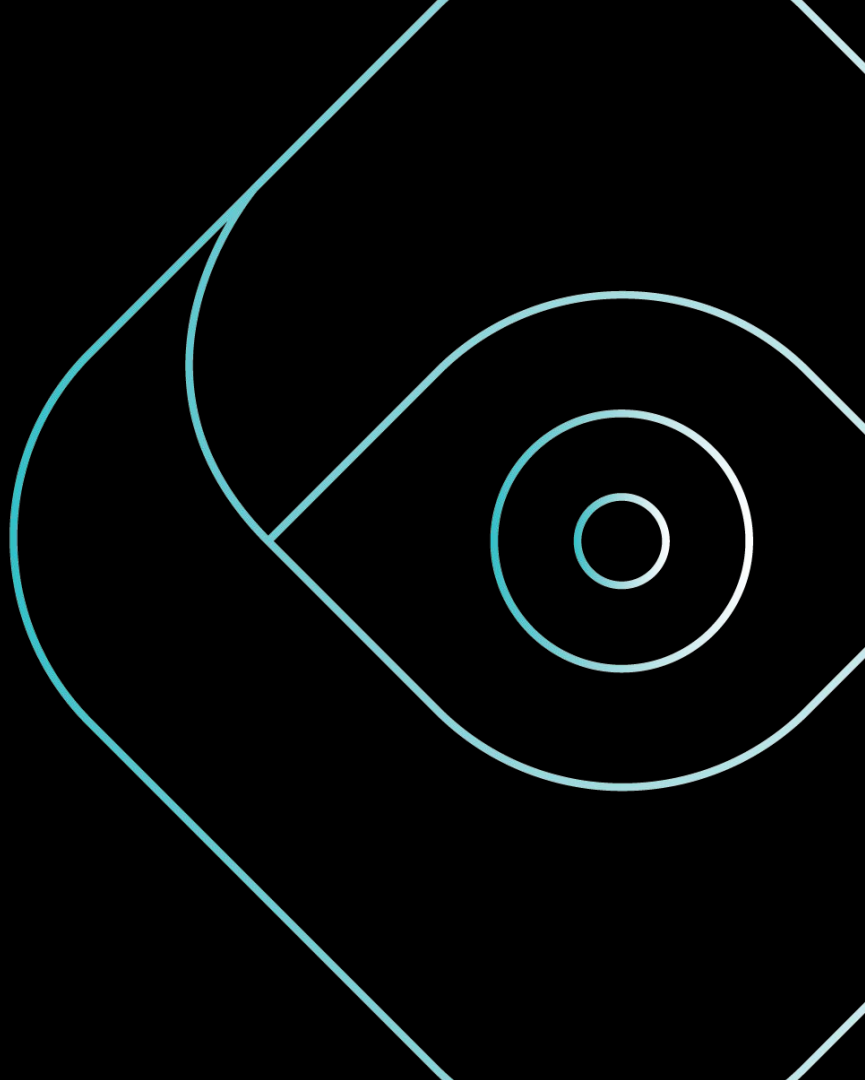
sending new values every second

Every.

Day.

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Chapter 3: Make something that people want to use!



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OQL (Oden Query Language)

Query: Daily Factory Performance Over Last 90 Days

TRADITIONAL PROCESS

User	Data Engineer + Domain Expert
Time	1 hour+

SQL

```
SELECT
  Day,
  i.factory_name,
  sum(output)/sum(target) as performance,
(WITH
  TIMESTAMP_SUB(CURRENT_TIME(), INTERVAL 90D) as start_time,
  CURRENT_TIME() as end_time,
  intersections as (
    SELECT * FROM (
      SELECT s.start_timestamp, s.type, null FROM states s
      WHERE s.start_timestamp BETWEEN start_time & end_time UNION ALL
      SELECT r.start_timestamp, null, r.product FROM runs r
      WHERE r.start_timestamp BETWEEN start_time & end_time
    ) INTERPOLATE (s.type as s.type, r.product as r.product)
  ),
  factory_lines as (
    SELECT id FROM lines WHERE factory_id = 'a82f9f08-1e97-4328-adfc-5869069c0091'
```

```
SELECT
  i.start_timestamp,
  TIMESTAMP_TRUNC(i.start_timestamp, DAY) as Day,
  i.factory_name,
  (avg(m.value) * (i.end_timestamp - i.start_timestamp)) as output,
  avg(t.target) as target,
FROM metrics m
ASOF JOIN intersections i ON m.line_id = s.line_id & m.timestamp >=
s.start_timestamp
JOIN targets t ON i.line_id = t.line_id & i.product = t.product
WHERE s.type = "Uptime" & m.id IN (
  select id from metric_definition where line_id IN (factory_lines) &
m.metricgroup_label = 'performance'
)
GROUP BY s.start_timestamp, i.factory_name)
GROUP BY Day, i.factory_name
```

- Requires database query language expertise
- Time-consuming trial & error
- Limited to few users

Complex systems lead to inaccurate & abandoned initiatives

ODEN QUERY LANGUAGE

User	Engineering & Operations
Time	10 – 30 Seconds

OQL

```
Aggregate aggregation.performance*100 as performance
where factory.id = ‘ a82f9f08-1e97-4328-adfc-5869069c0091 ‘
& time from now () – 90D to now()
group by factory.name, bucket 1D
```

- No-code query exercise for user
- Immediate insights lead to adaptable query
- Ops-centric view without process engineer

Oden provides the most flexible real-time adaptable platform for the manufacturing ecosystem

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Oden Product Suite

Factory Analytics
Reporting +
Dashboards
Operator Scorecards
Root-Cause Analysis

All Operations Mgmt

Process AI
Real-time process recommendations for Operators

Uptime

Knowledge AI
Intelligent work instructions for Operators

Planned Downtime

Anomaly Detection
Intelligent alerts for all types of process deviations

Unplanned Downtime

Forge
LLM-driven AI assistant for all configurations, data analysis, and insight generation

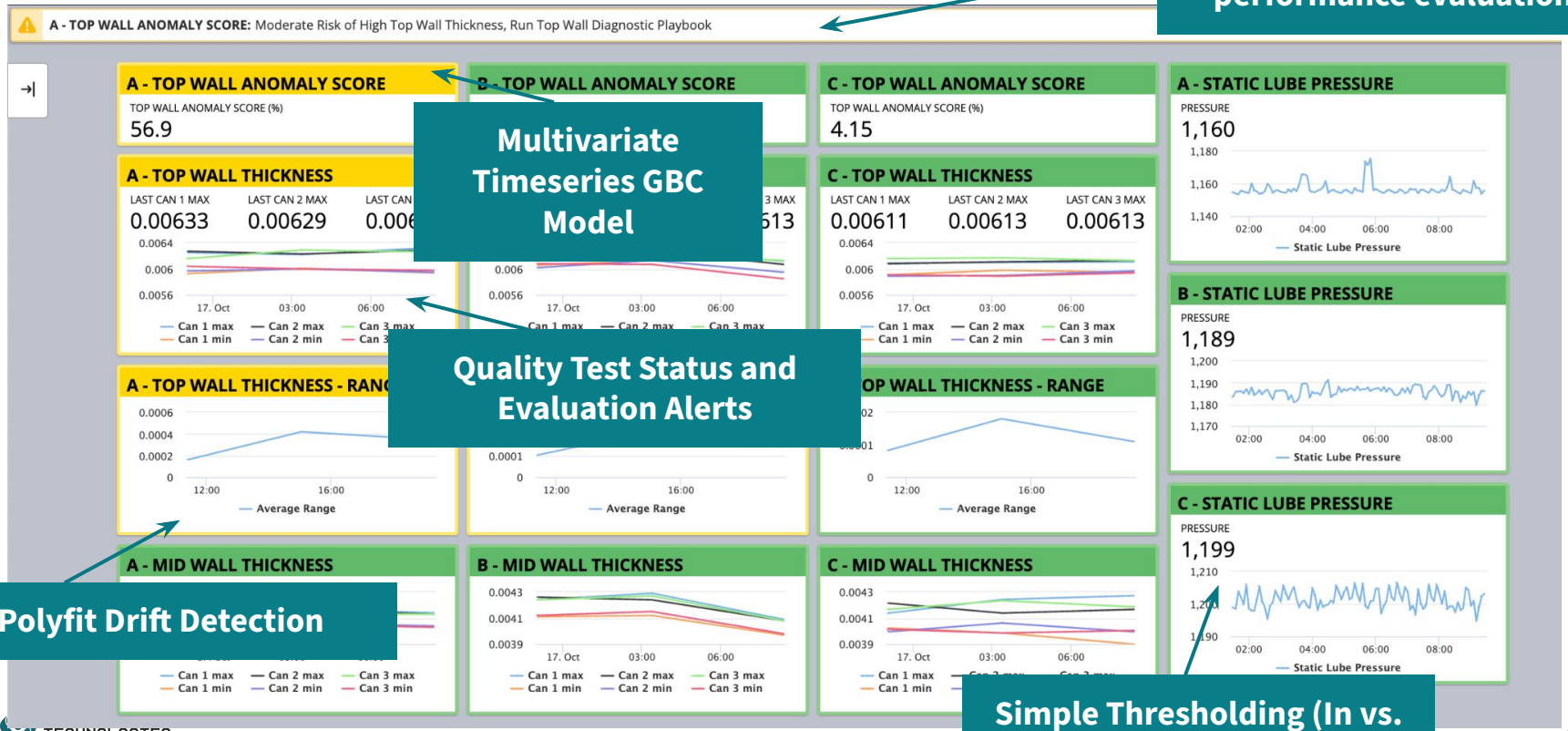
Data Engine
Data cleansing, contextualization, inferencing & enrichment

Data Connect
Integrate data from all types of software systems

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ProcessAI

High Level alerting Automatically converted into data labels for model performance evaluation



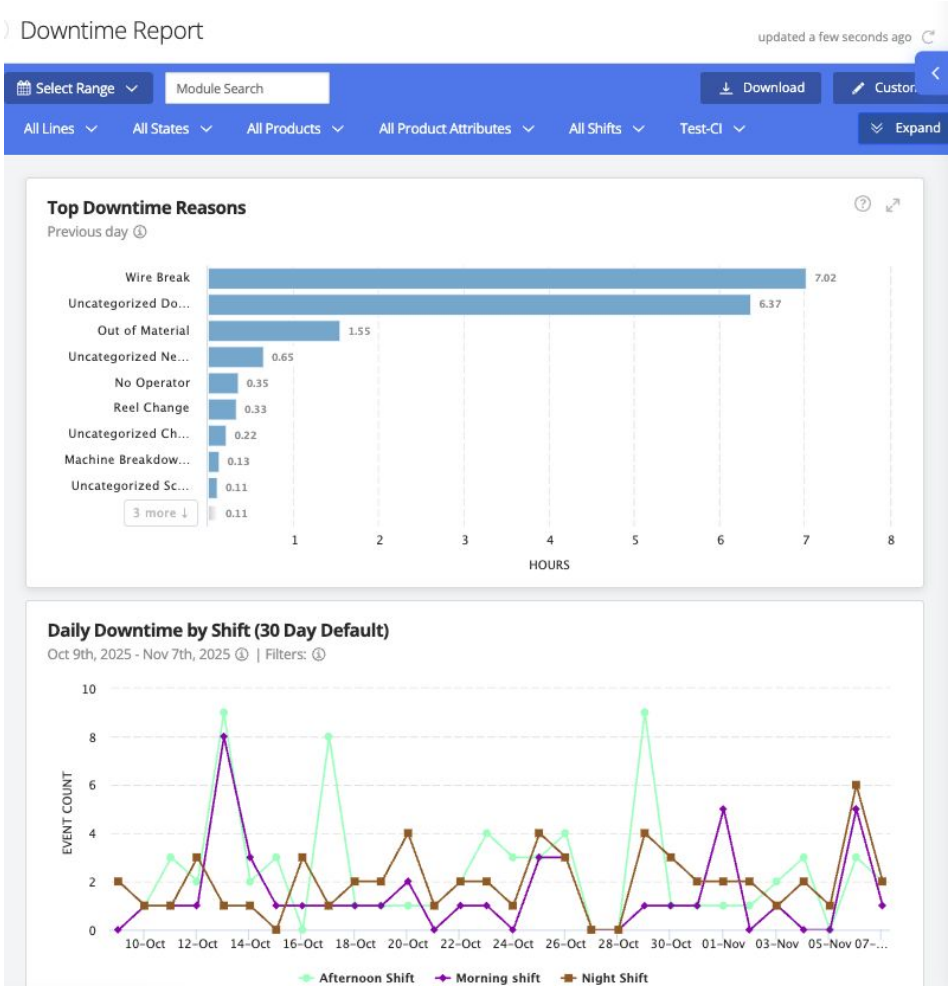
Multivariate Timeseries GBC Model

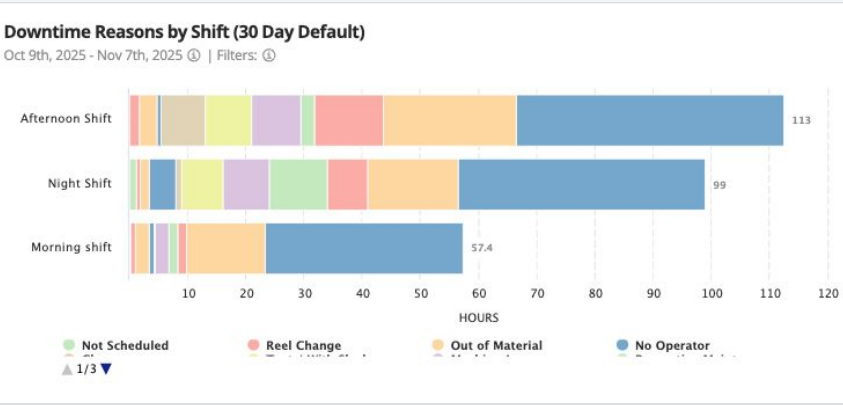
Quality Test Status and Evaluation Alerts

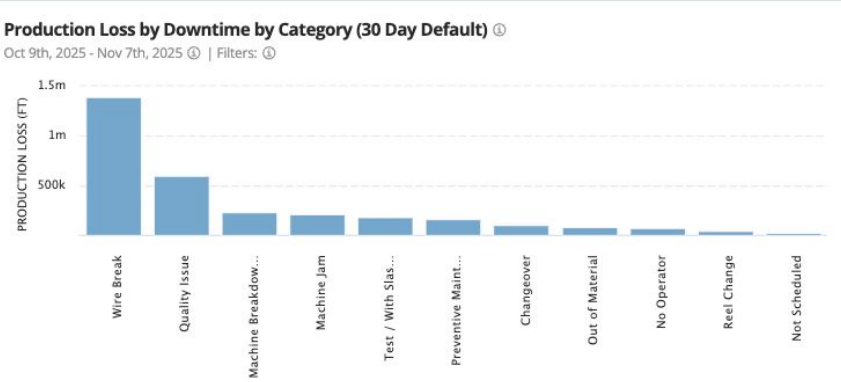
Polyfit Drift Detection

Simple Thresholding (In vs. Out of Range)


Now
Dashboards
Explore
Process AI
Forge
OQL
Data Status
GRADE CHANGE
MOISTURE
Z PLY
COST
Read-Only Mode
MATERIAL FLOW 2D
CURRENT
316.79
RECOMMENDED
316.33
LINE SPEED (FPM)
CURRENT
380.21
RECOMMENDED
399.79
REFINER FLOW 2D
CURRENT
340.73
RECOMMENDED
406.06
REFINER FLOW 2E
CURRENT
285.30
RECOMMENDED
309.65
STEAM FLOW 1A
CURRENT
34.39
RECOMMENDED
31.49
PAPER STRENGTH Z PLY
OFFLINE
80.2
PREDICTED
82.7
TARGET
72.0
83.25
83
82.75
82.5
82.25
19:00
19:30
20:00
20:30
Predicted
MOISTURE CONTENT
OFFLINE
5.102
PREDICTED
5.051
TARGET
5
5.0512
5.0512
5.0511
19:00
19:30
20:00
20:30
Predicted
PAPER STRENGTH TENSILE
OFFLINE
148
PREDICTED
145
TARGET
132
145.5
145
144.5
19:00
19:30
20:00
20:30
Predicted
Feedback
Data Entry
Operator Log
CURRENT COST ($/FT)
77.22
PREDICTED COST ($/FT)
77.111
PREDICTED SAVINGS ($/HR)
6.556
Get Rec

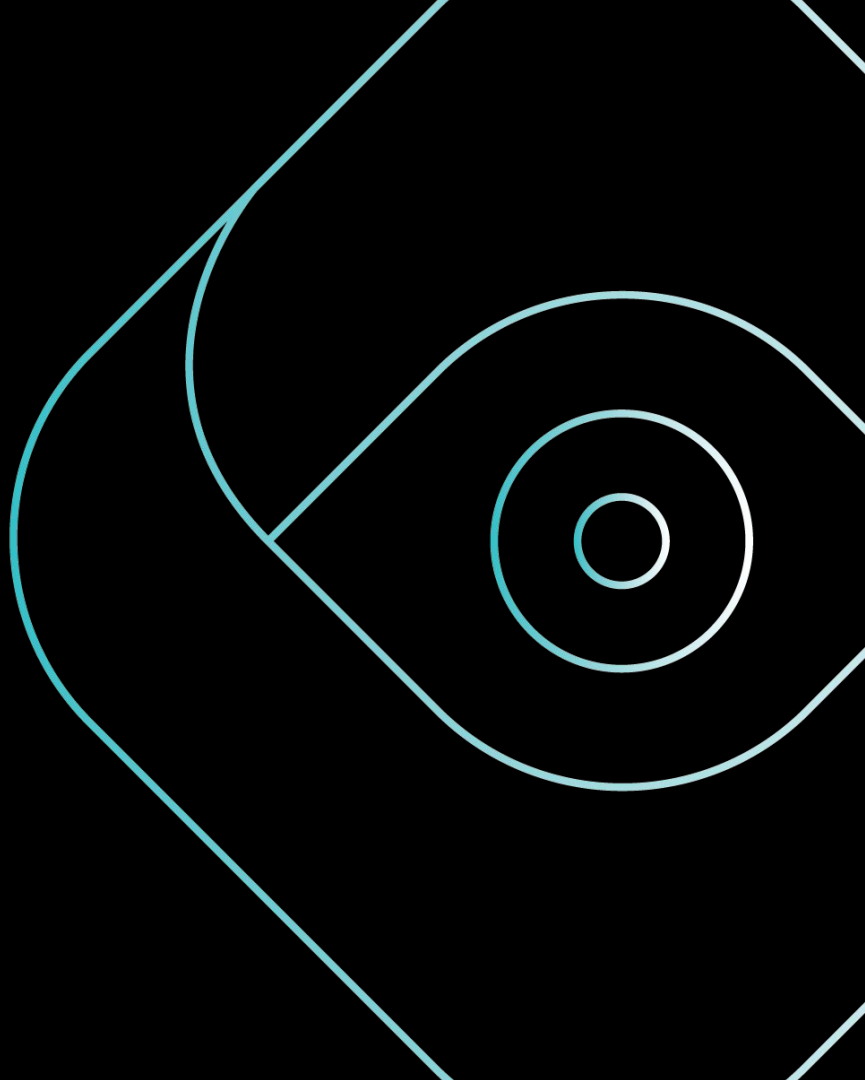

Downtime Report
updated a few seconds ago
Select Range
Module Search
Download
All Lines
All States
All Products
All Product Attributes
All Shifts
Test-CI
Expand
Top Downtime Reasons
Previous day
Wire Break
Uncategorized Do...
Out of Material
Uncategorized Ne...
No Operator
Reel Change
Uncategorized Ch...
Machine Breakdow...
Uncategorized Sc...
3 more
7.02
6.37
1.55
0.65
0.35
0.33
0.22
0.13
0.11
0.11
HOURS
Daily Downtime by Shift (30 Day Default)
Oct 9th, 2025 - Nov 7th, 2025 | Filters:
EVENT COUNT
Afternoon Shift
Morning shift
Night Shift


Downtime Reasons by Shift (30 Day Default)
Oct 9th, 2025 - Nov 7th, 2025 | Filters:
Afternoon Shift
Night Shift
Morning shift
113
99
57.4
HOURS
Not Scheduled
Reel Change
Out of Material
No Operator
1/3


Production Loss by Downtime by Category (30 Day Default)
Oct 9th, 2025 - Nov 7th, 2025 | Filters:
PRODUCTION LOSS (FT)
Wire Break
Quality Issue
Machine Breakdow...
Machine Jam
Test / With Slas...
Preventive Maint...
Changeover
Out of Material
No Operator
Reel Change
Not Scheduled

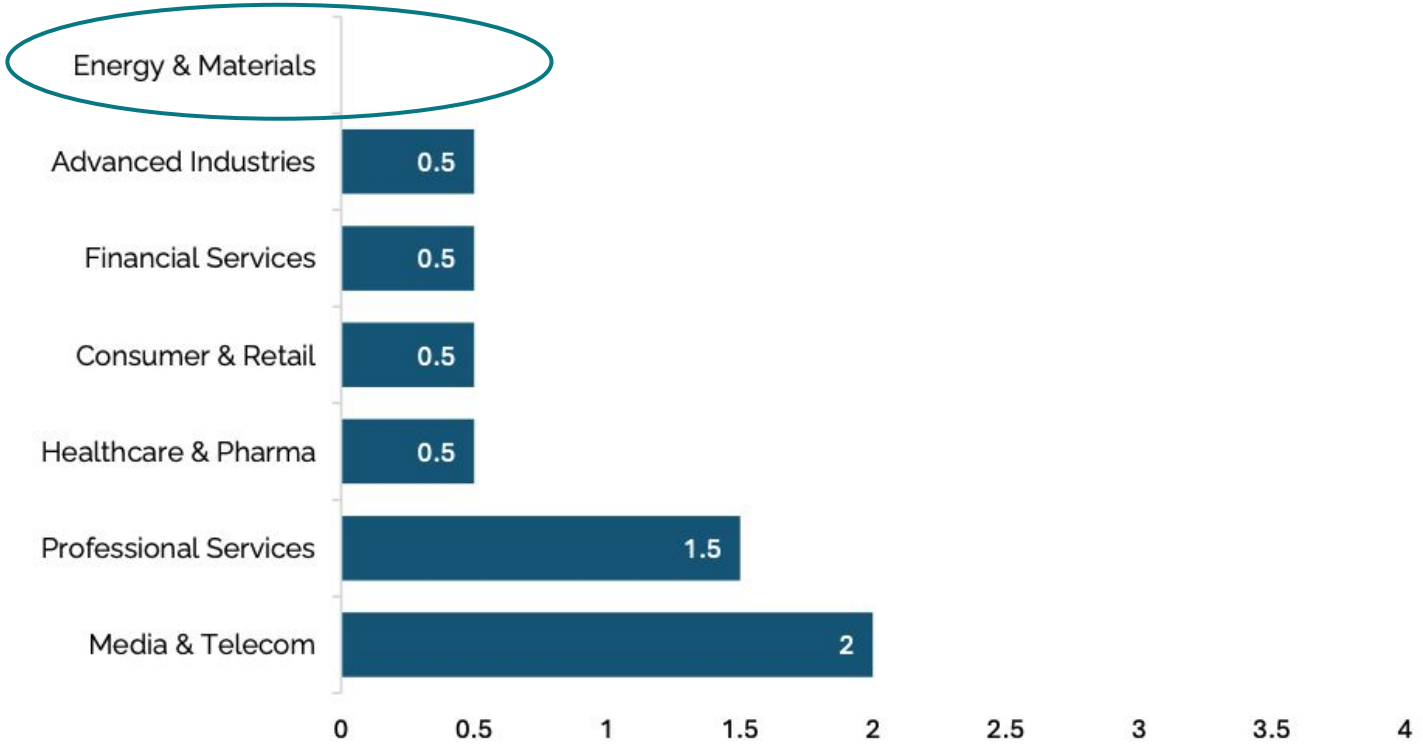
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Chapter 4: GenAI? In Manufacturing?

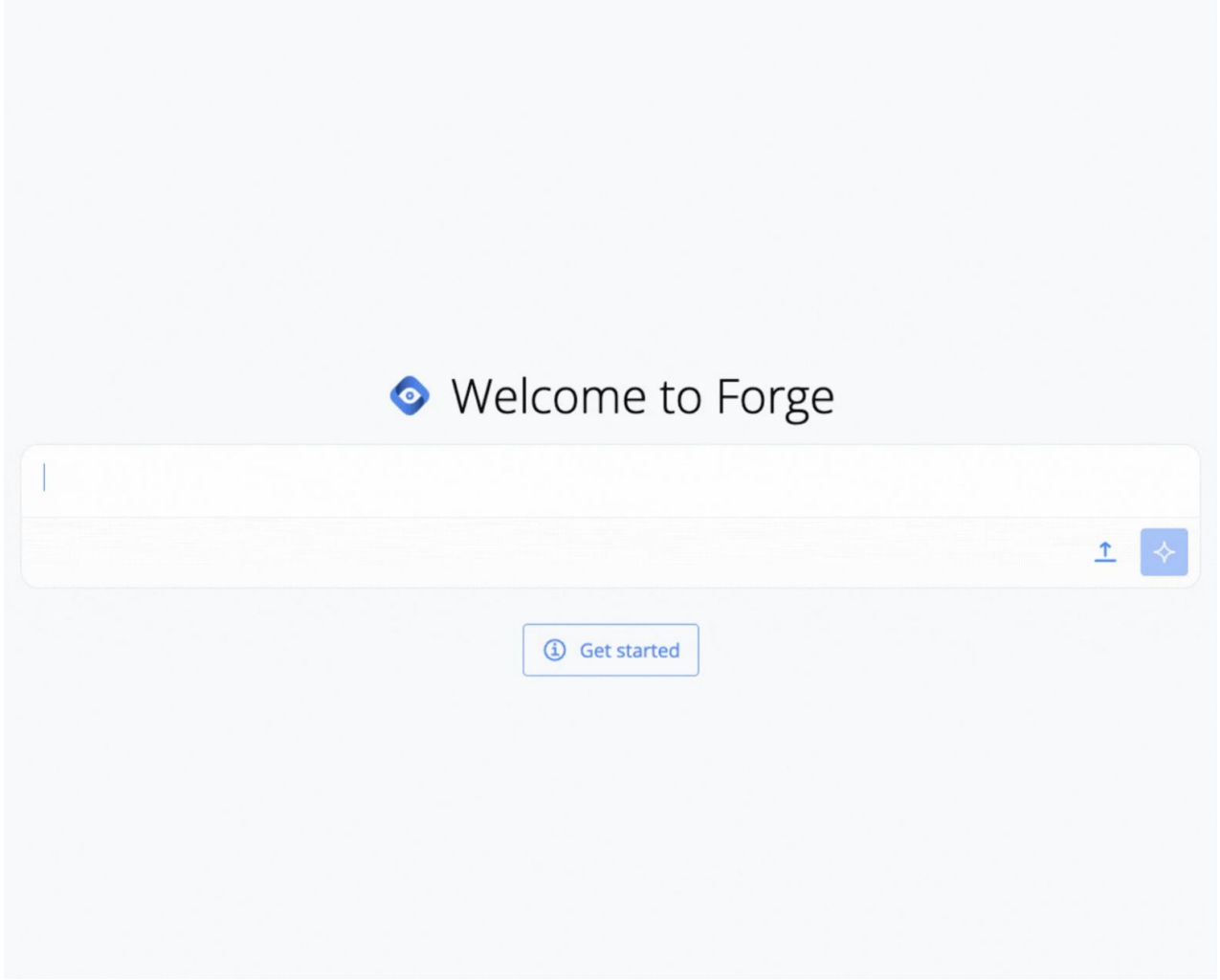


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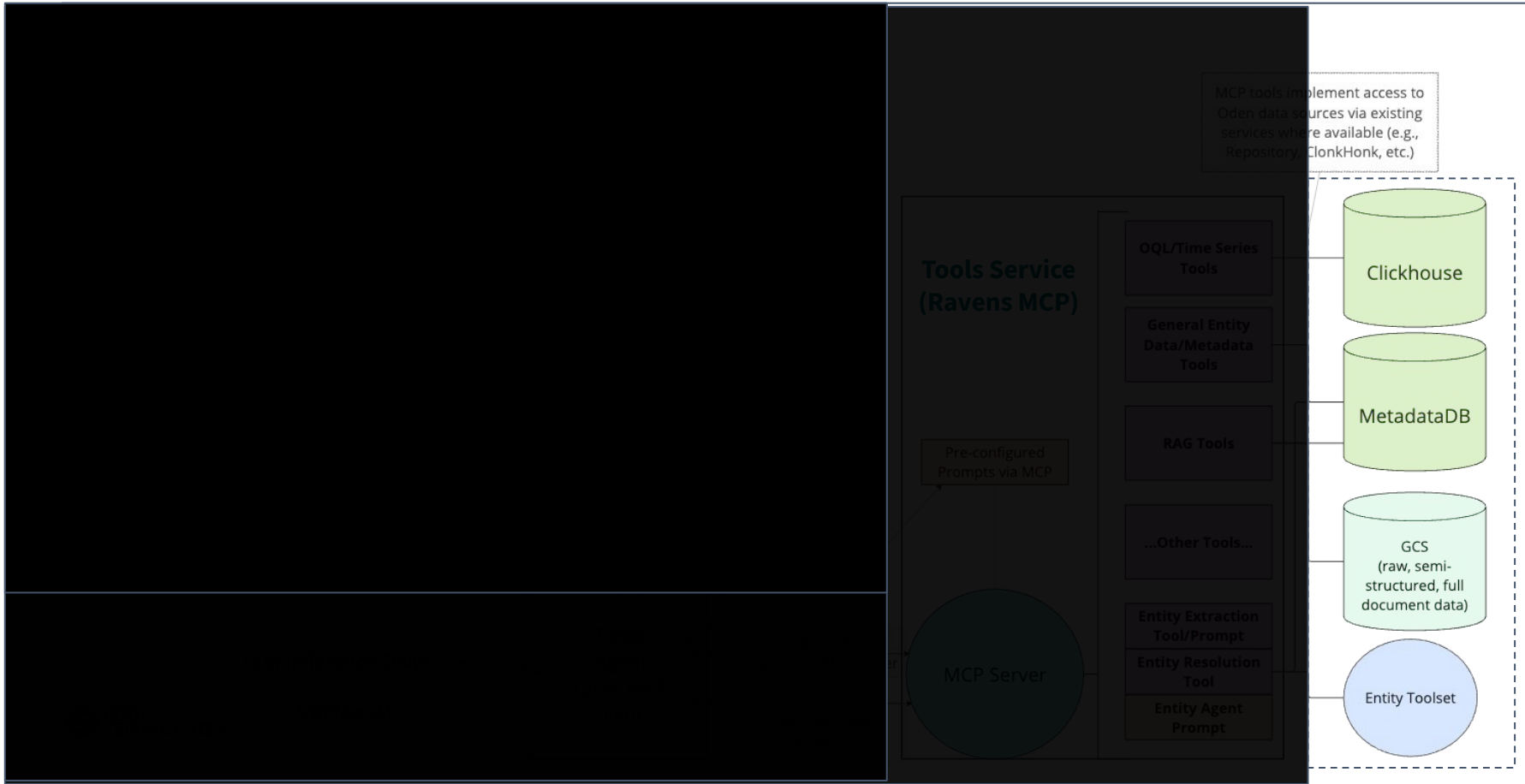
Exhibit: GenAI disruption varies sharply by industry



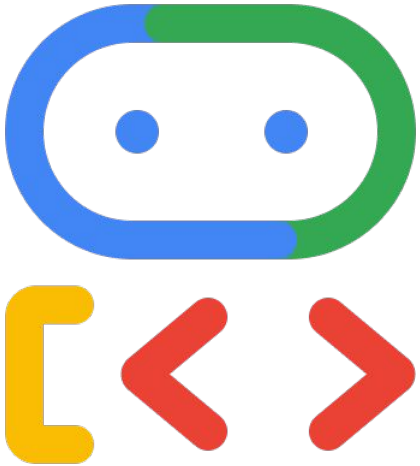
Sources: MIT The GenAI Divide STATE OF AI IN BUSINESS 2025

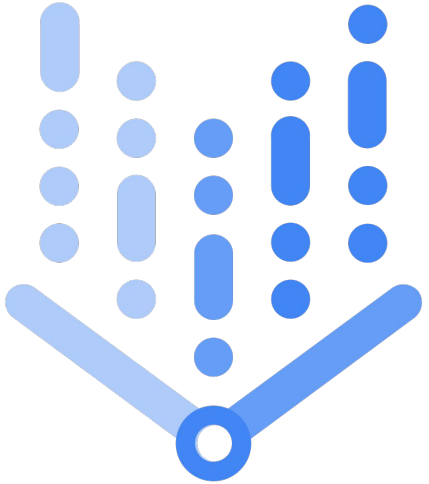

Welcome to Forge
Get started

Detailed Forge Architecture
(Model Context Protocol, Google ADK)



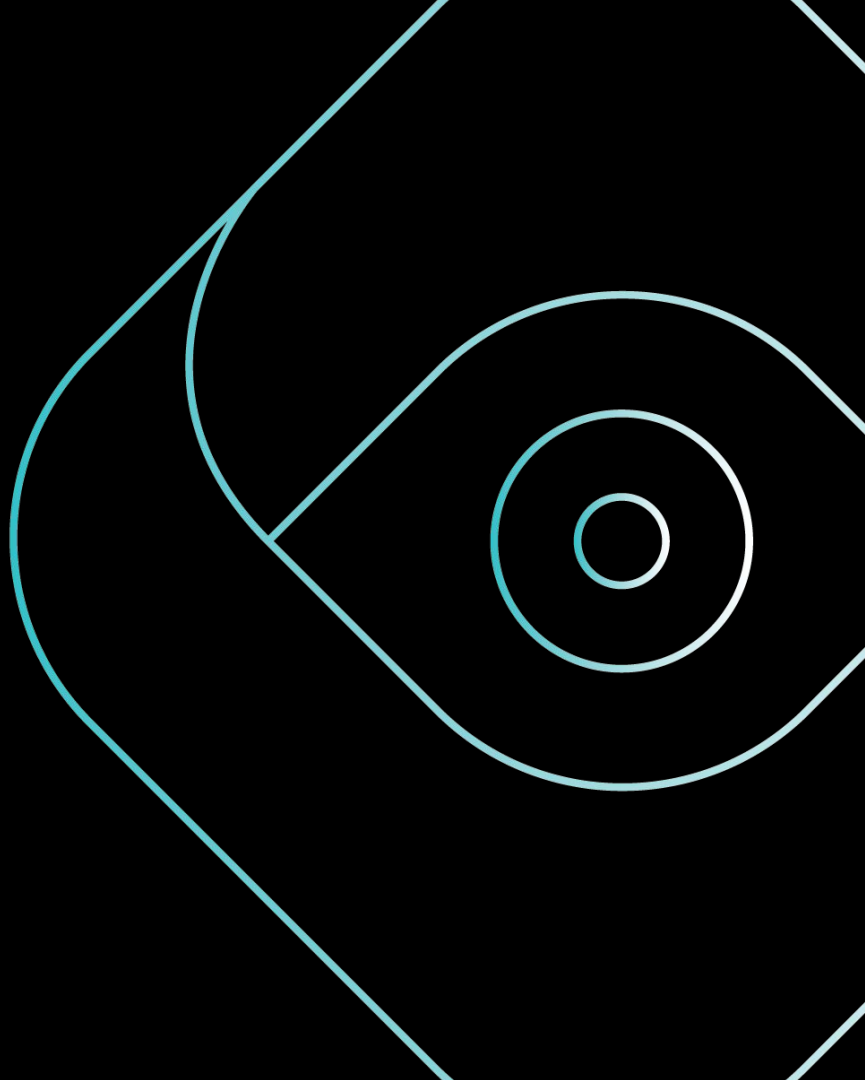
ADK for Agent Development, Vertex AI for Deployment





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Chapter 5: Launch!



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Model Evals - Is this just testing all over again?

Purpose:

- LLMs are non-deterministic & a bit mysterious at times
  - Need systematic, repeatable ways of measuring their output quality
  - Ensure consistent and safe responses in production LLMS

How evals work:

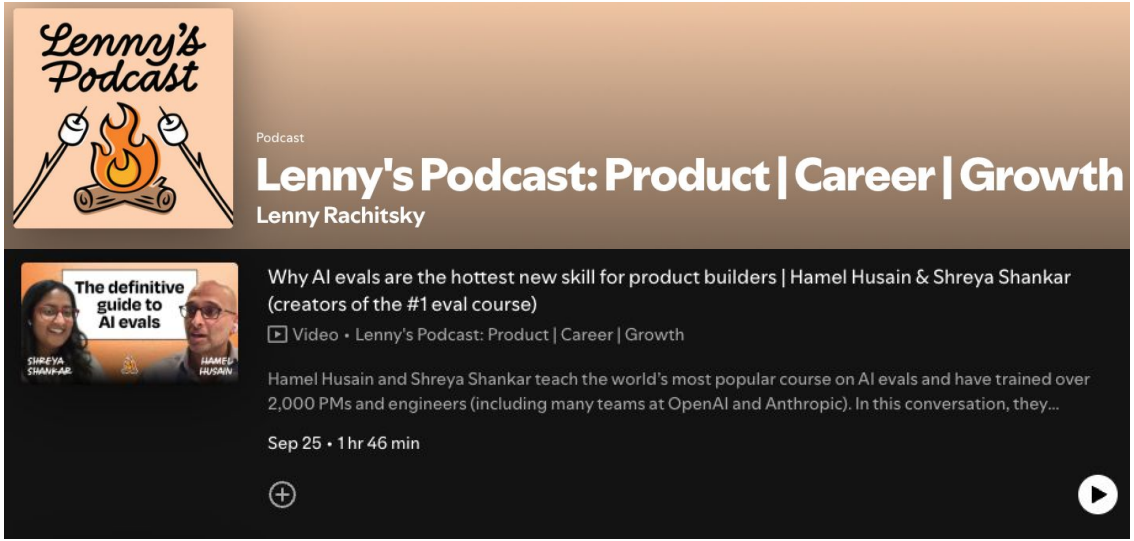
- Metrics like “correctness”, “relevance”, “helpfulness”, “safety”, “factuality”
- Mixture of Human annotation and LLM-as-judge evaluation
- Some traditional unit tests/checks as well, formatting, etc.

Abundance of options:

- Braintrust, VertexAI, Langfuse, MLFlow etc.
- Advice: avoid decision paralysis, demo and pick any of them



More on Model Evals



https://langfuse.com/docs

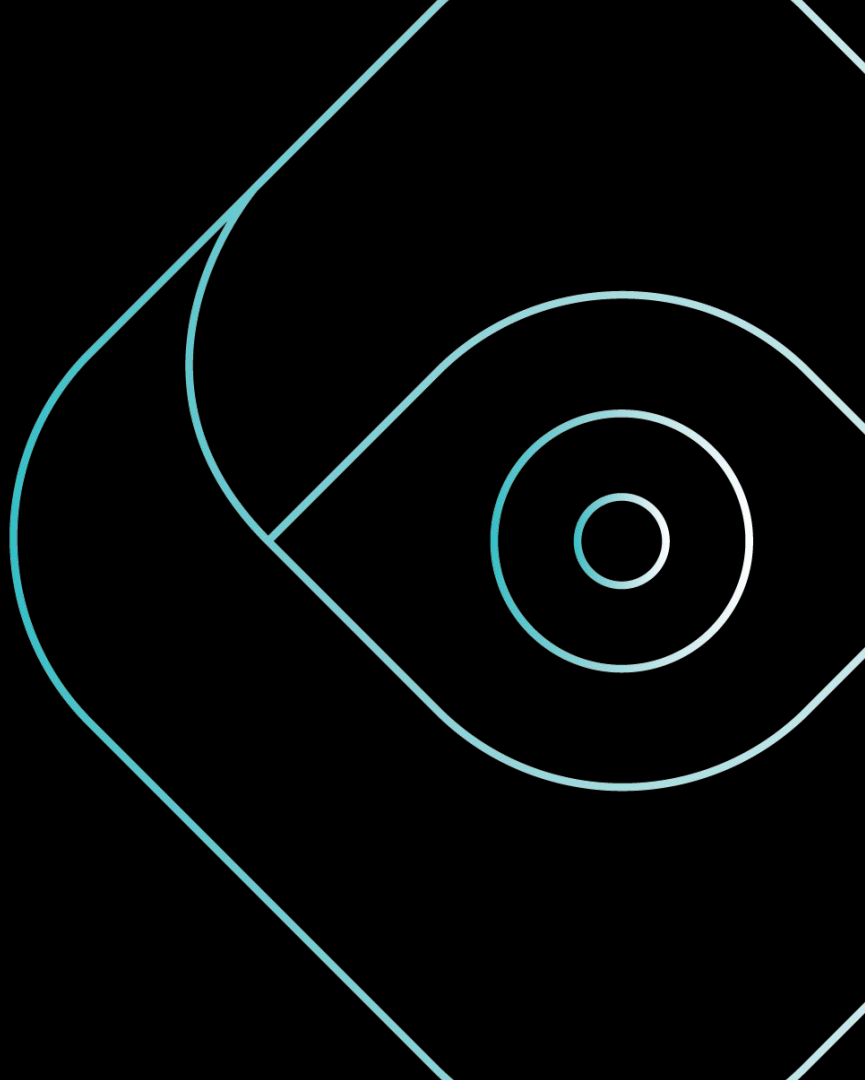
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Thank You!

I hope you learned something new today!

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joseppy.ca



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ODEN TECHNOLOGIES

Thank You!

I hope you learned something new today!

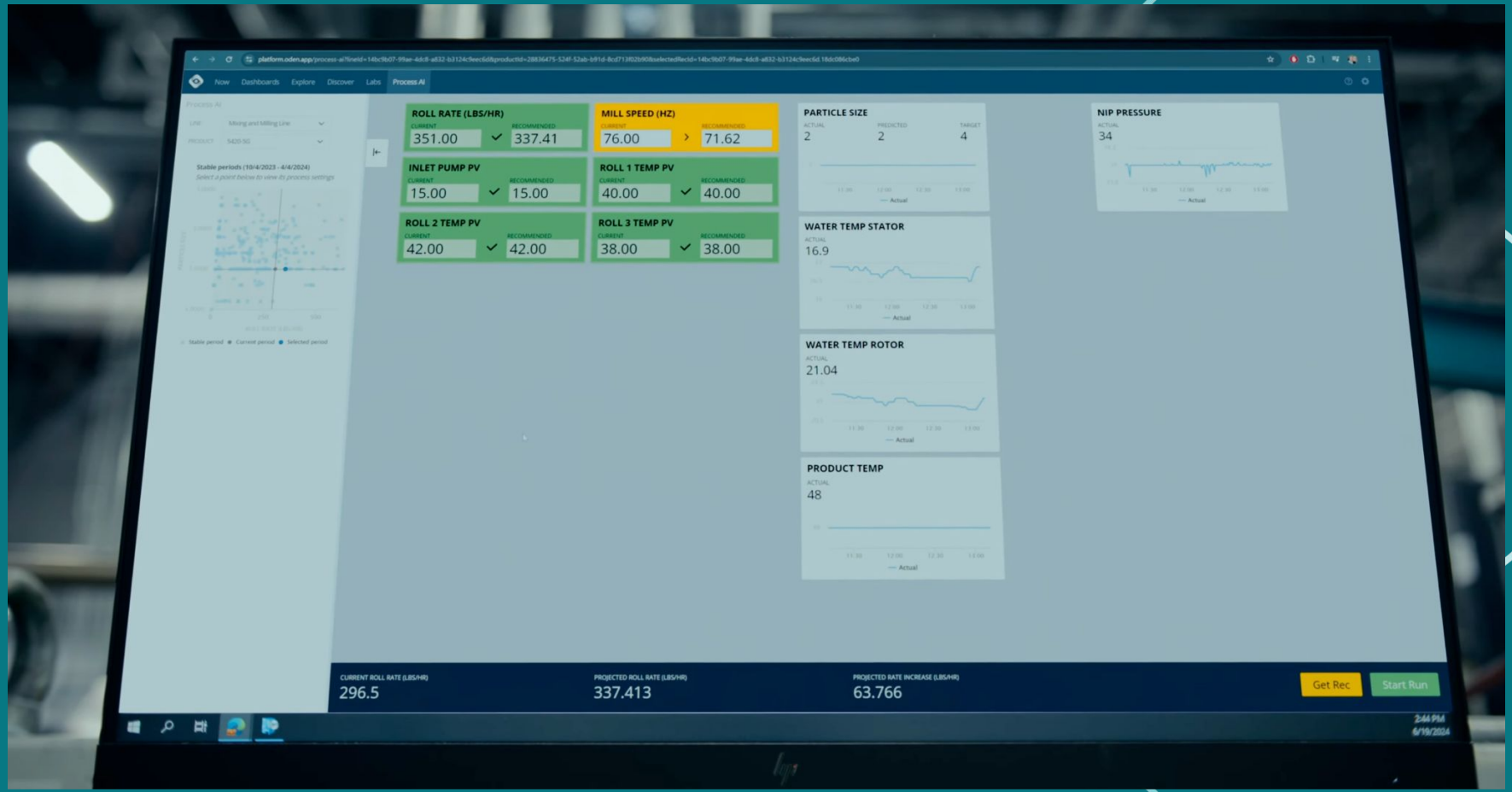
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joseppy.ca

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Empower the Front-Line Real-Time AI Recommendations

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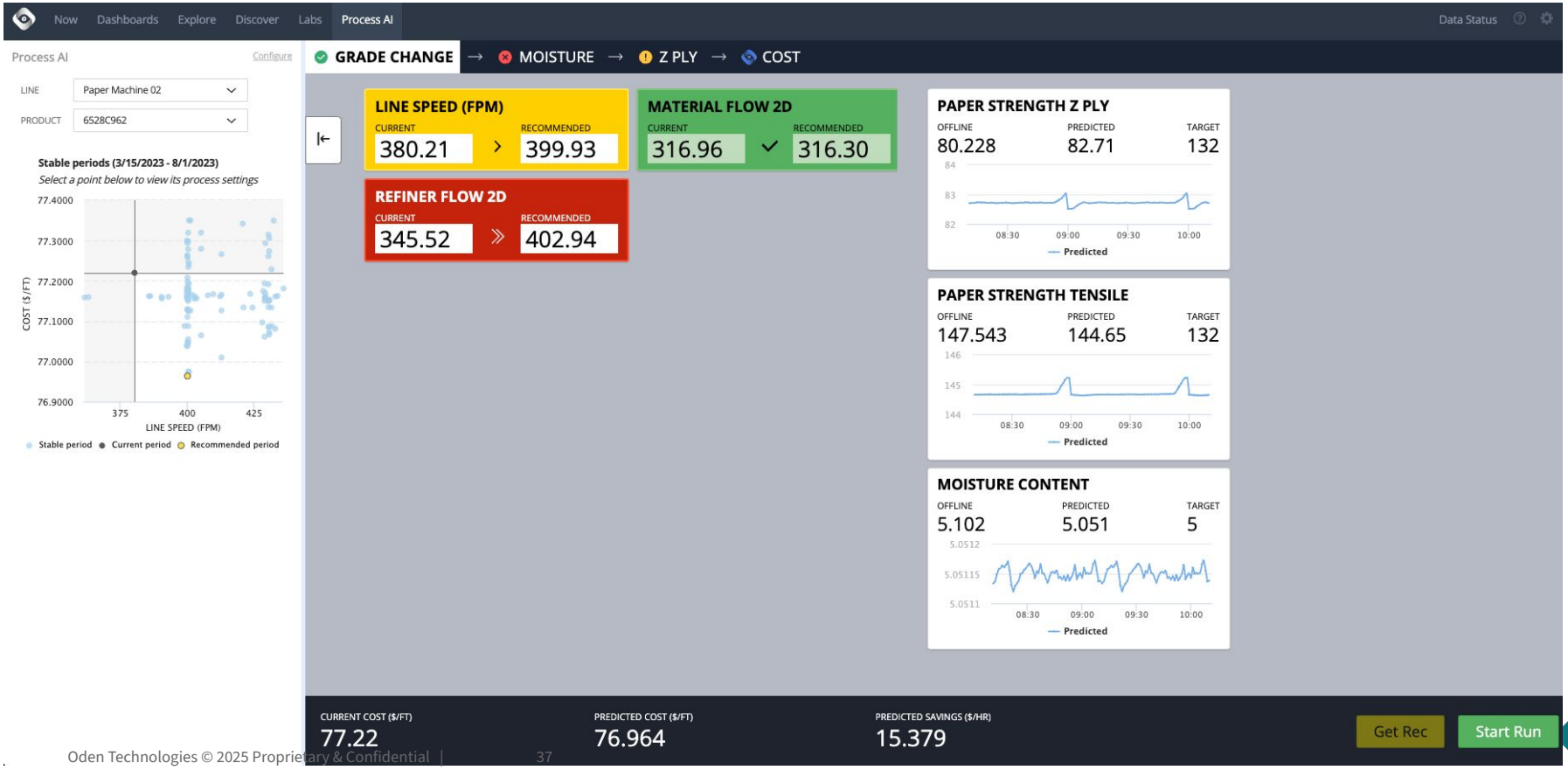
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Process AI: Real-Time Operator Recommendations

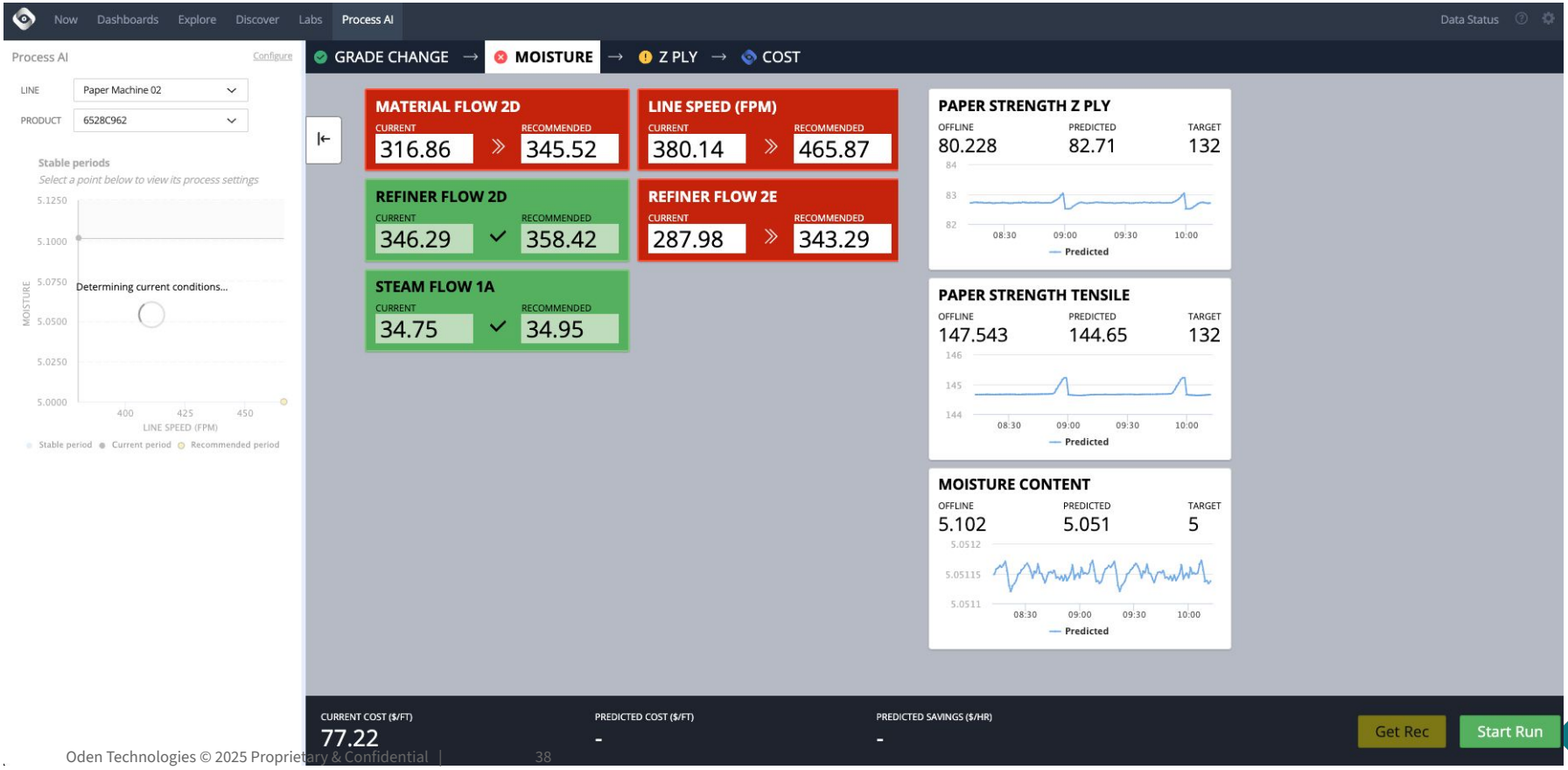


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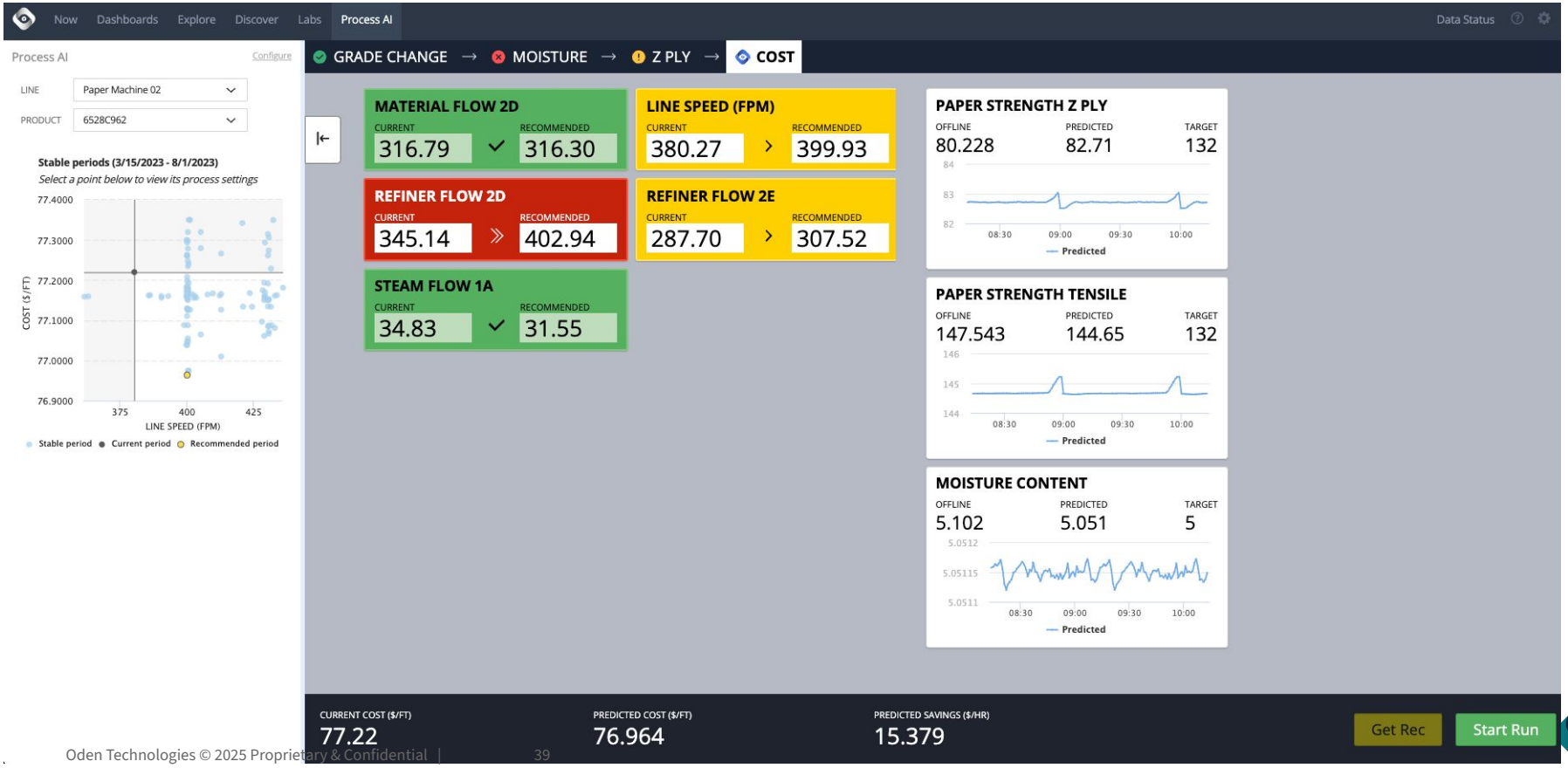
Process AI: Incremental Recommendations for Paper



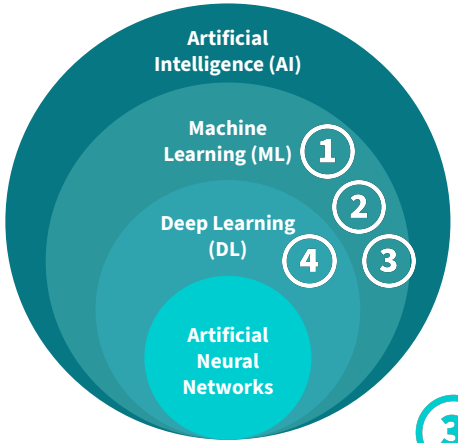
Process AI: Incremental Recommendations for Paper



Process AI: Incremental Recommendations for Paper



AI/ML Techniques used in the Oden Product Suite



1 Linear Models

Uses: Predictive Quality
Benefits: Interpretability, fast, easy to train, scalable

Oden uses a variety of different approaches (LASSO, Elastic Net, PCA-Regression) to predict quality metrics. These are powerful as all of the model variables can be visualized and trust is built.

Oden also uses Deep Learning and other models for PQ problems to validate results.

2 Tree Models and Ensembles

Uses: Predictive Quality, Root Cause Analysis
Benefits: Interpretability, non-linear separations, handling categorical data

Used to identify key contributors separating periods of interest from baselines, i.e. changeover detection

3 Optimization Models

Uses: Process AI Recommendations
Benefits: Solve complex multi-objective multi-constraint problems

Oden uses optimization models to provide process setting recommendations directly to operators in real-time.

4 Deep Learning and Large Language Models

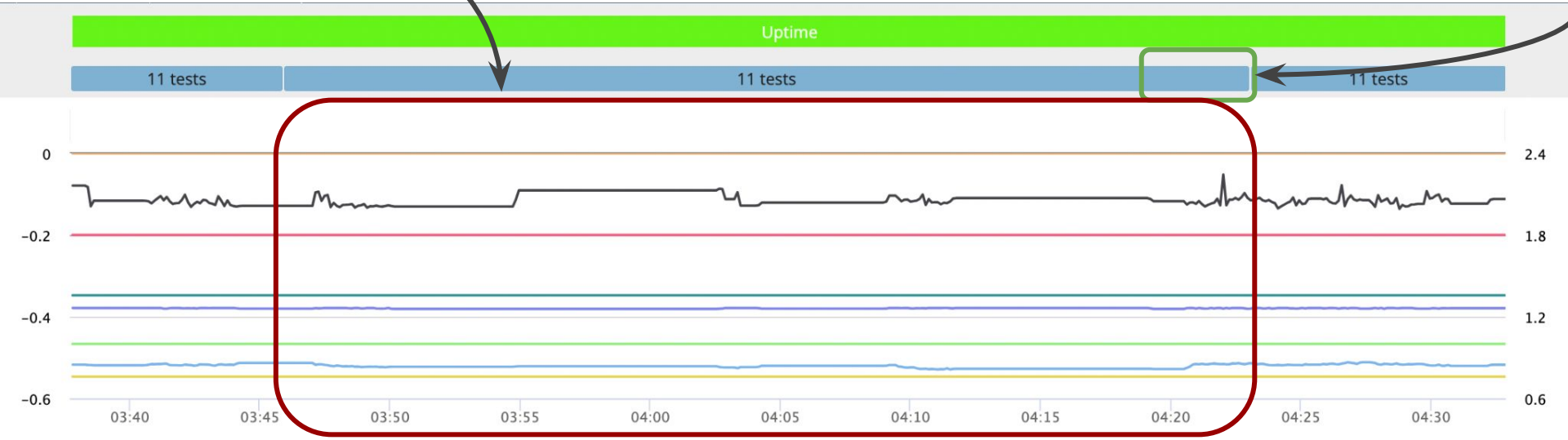
Uses: Stable Periods, Knowledge AI [in development]
Benefits: Powerful learning

Oden uses Deep Learning models for Stable Period analysis for Process AI, as well as in exploratory work with Knowledge AI and providing work instructions.

Problem Statement: Predictive Quality

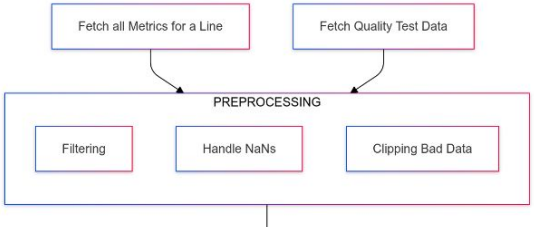
Use **multi-dimensional time series** signatures:

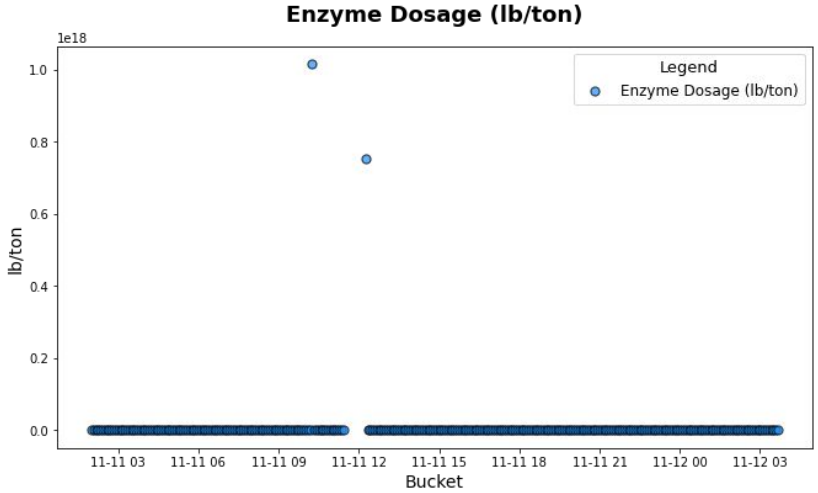
To predict a single quality test **label** at the end of each run:

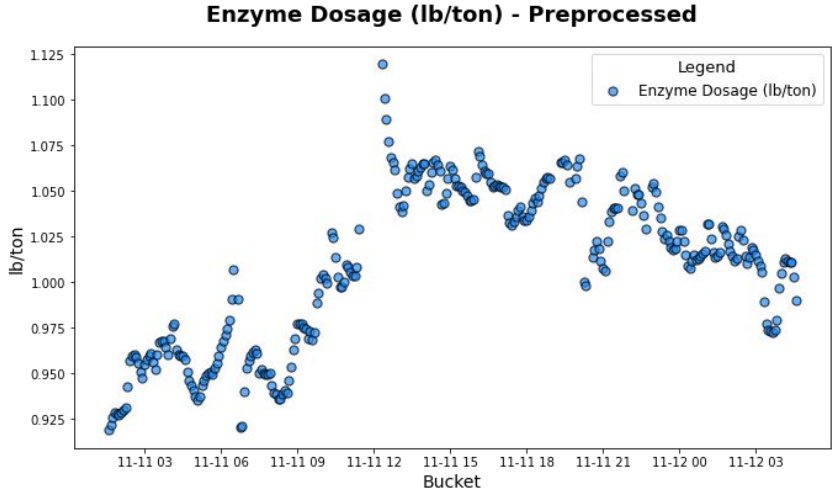


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Example PQ Pipeline - Preprocessing

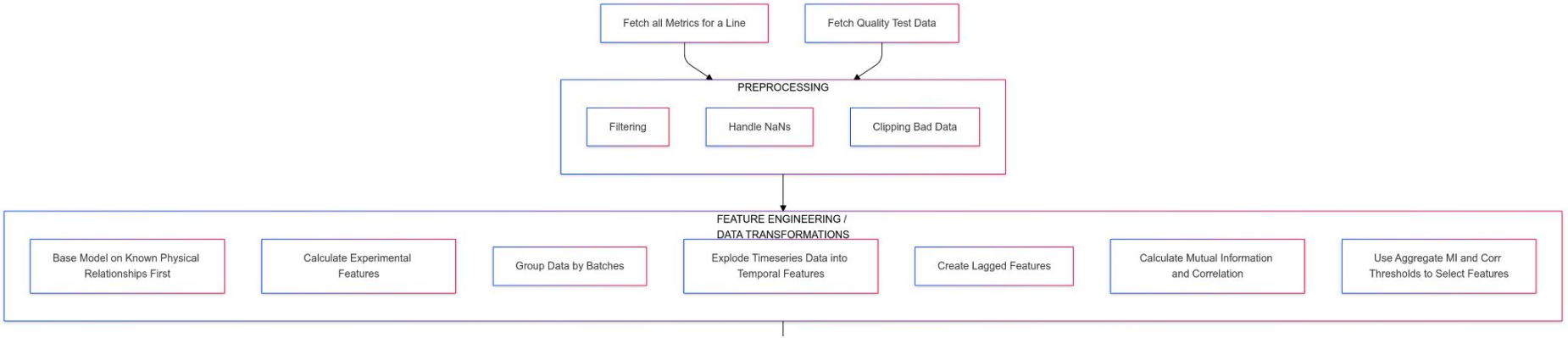






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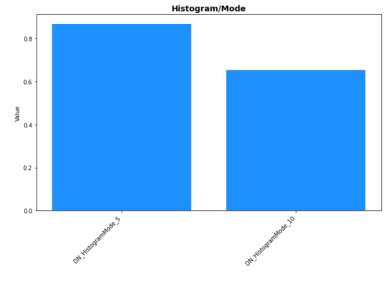
Example PQ Pipeline - Feature Engineering

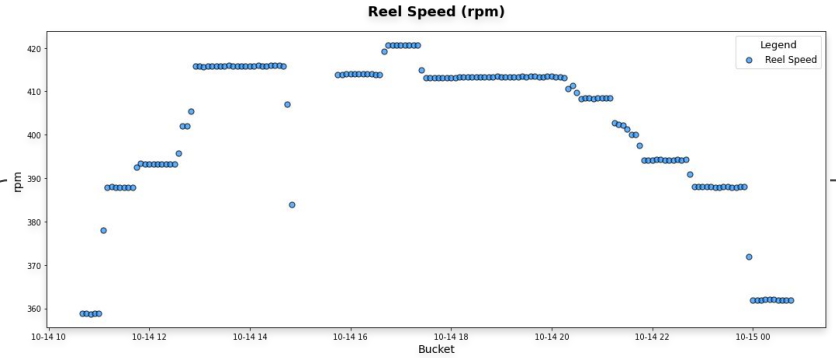


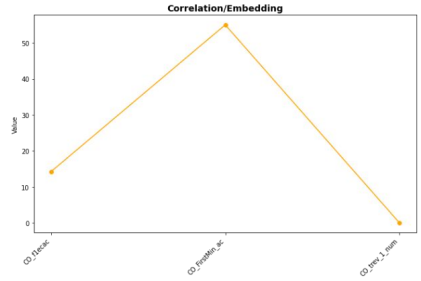
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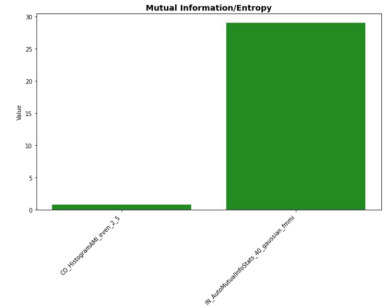
Example PQ Pipeline - Feature Engineering

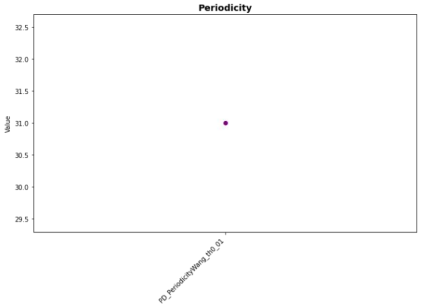
Time Series Decomposition into Core Stats

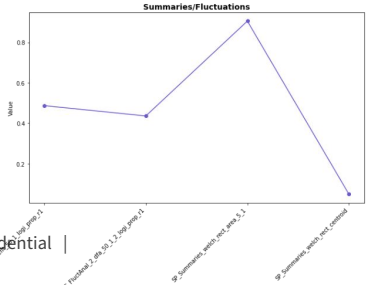


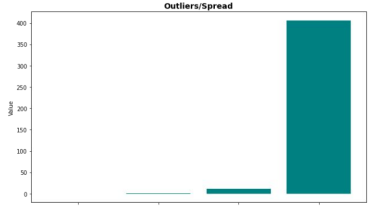


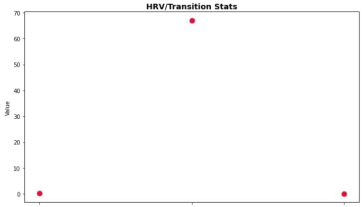












Source: catch22: CAnonical Time-series CHaracteristics
https://link.springer.com/article/10.1007/s10618-019-00647-x

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Feature Selection and the Lag Effect

A change in one metric may not impact output quality for N minutes.

How can we find N?

If available, ask a **Domain Expert!**

Otherwise, we can estimate this with lagged features, and ranking metrics (e.g. corr)



bucket	cd821ae1-c69b-5d37-b751-84c28bcf14e9	cd821ae1-c69b-5d37-b751-84c28bcf14e9_lag_5min	cd821ae1-c69b-5d37-b751-84c28bcf14e9_lag_20min	cd821ae1-c69b-5d37-b751-84c28bcf14e9_lag_60min
2024-04-09 19:30:00+00:00	2.053912	1.997217	2.514325	2.594364
2024-04-09 19:35:00+00:00	2.027539	2.065962	2.516855	2.779457
2024-04-09 19:40:00+00:00	2.014745	2.091880	2.469408	2.942021
2024-04-09 19:45:00+00:00	2.015056	2.080691	2.381821	2.937846
2024-04-09 19:50:00+00:00	1.985830	2.084497	2.358898	2.894803
...	...	...	...	...
2024-11-14 23:35:00+00:00	3.318202	3.758257	3.936180	3.781135
2024-11-14 23:40:00+00:00	3.298853	3.529197	3.918642	3.784988
2024-11-14 23:45:00+00:00	3.275909	3.468211	3.874753	3.764397
2024-11-14 23:50:00+00:00	3.304411	3.396445	3.864001	3.851068
[illegible]	3.321725	3.347016	3.862047	3.946134

31806 rows × 4 columns

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Feature Selection and Ranking Metrics

Feature Space:

-> 100's to 1000's of raw & transformed metrics

Systematic Filtering Methods:

1. Human-In-The-Loop Feedback (Physics-Based/ State-Space Modelling)

2. Mutual Information/ Absolute Correlation Scores

3. Concentration of Noise/"Bad Data", nunique(), etc.

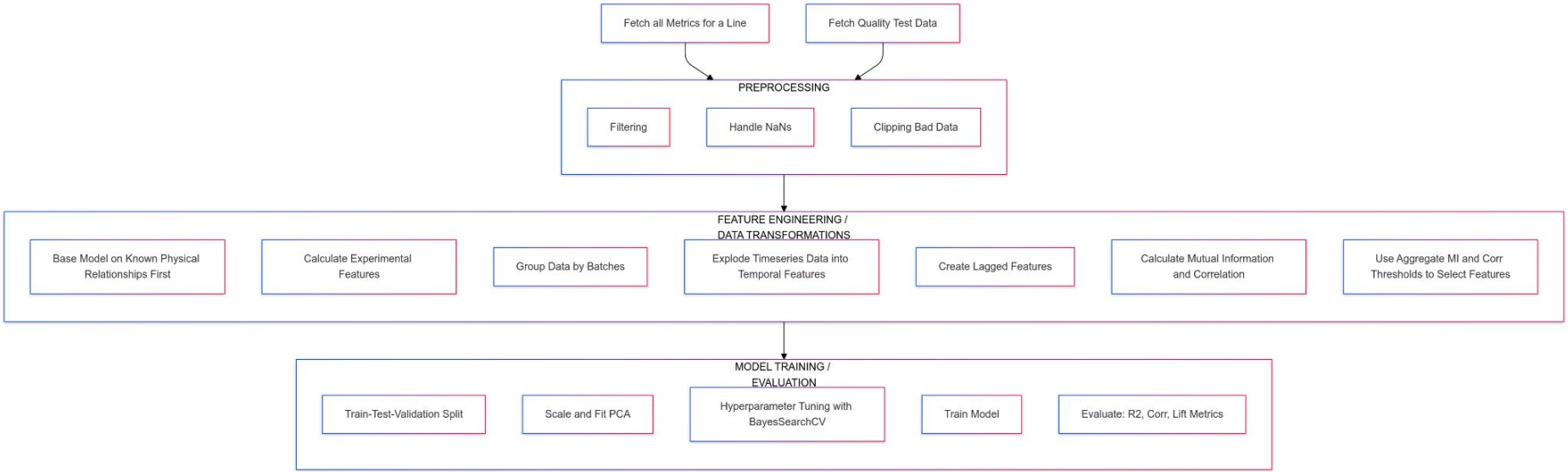
metric	mutual_information	abs_correlation	unique_values
10119216-9108-48b8-835b-29a83ae4bc5f-CO_Histog...	0.007816	0.020257	352
671b338c-b982-4dbe-adb0-393126f8f64c-CO_FirstM...	0.028868	0.082565	11
b6c70614-6e25-4a7b-86f7-a02702fedcf7-FC_LocalS...	0.028162	0.037197	10
d6786dea-bf97-43d0-83d3-60d9e1b0c88e-DN_Histog...	0.000168	0.004669	1378
experiment_7-FC_LocalSimple_mean3_stderr	0.000000	0.002482	1378
...	...	...	...
306f4d01-8f33-5c89-ad05-11bf51b3b050-MD_hrv_cl...	0.000000	0.022463	20
1d2e853d-f2e5-58e2-af68-baaf57d4fa12-CO_Embed2...	0.012775	0.006904	1378
e2644829-1edb-50e9-8f20-4a3c97788227-DN_Spread...	0.000000	0.034116	1378
b3a5ca35-34e2-5eaf-83df-85617f630f0e-SB_Binary...	0.000603	0.080383	10
b6c70614-6e25-4a7b-86f7-a02702fedcf7-CO_Histog...	0.000563	0.005759	327

$$I(X,Y) = -\sum_{x_i, y_i} p(x_i, y_i) \log_2 \frac{p(x_i, y_i)}{p(x_i)p(y_i)}.$$

```
-> from sklearn.feature_selection import
mutual_info_regression
-> pd.Series.corr(labels)
-> pd.Series.nunique()
```

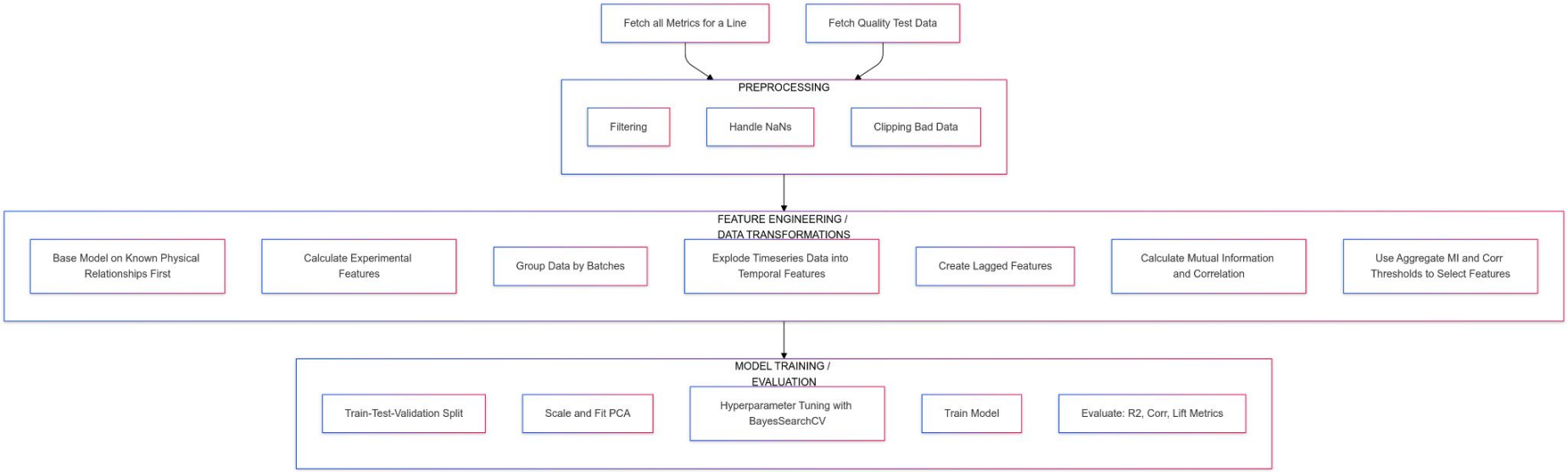
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Example PQ Pipeline - Train/Deploy



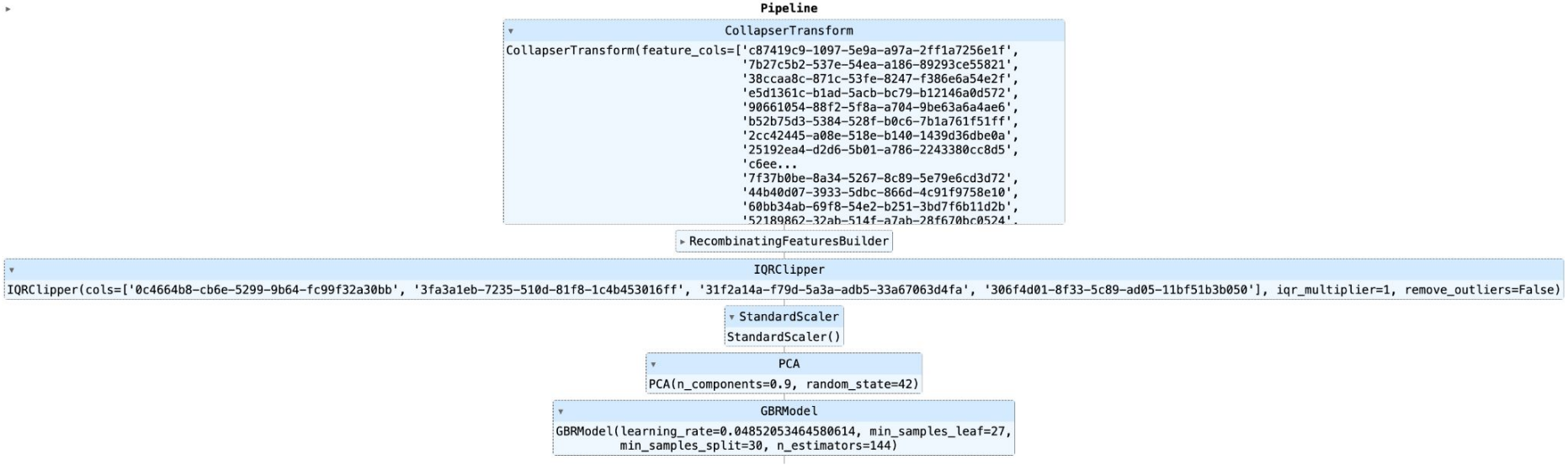
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Example PQ Pipeline - Train/Deploy



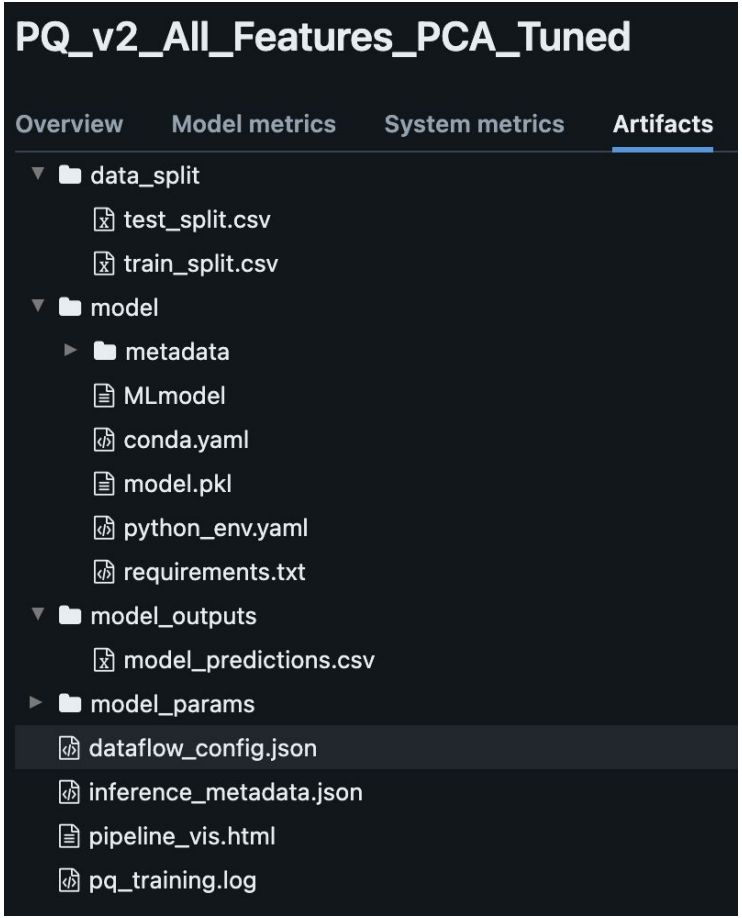
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Example Pipeline - Registered to MLFlow



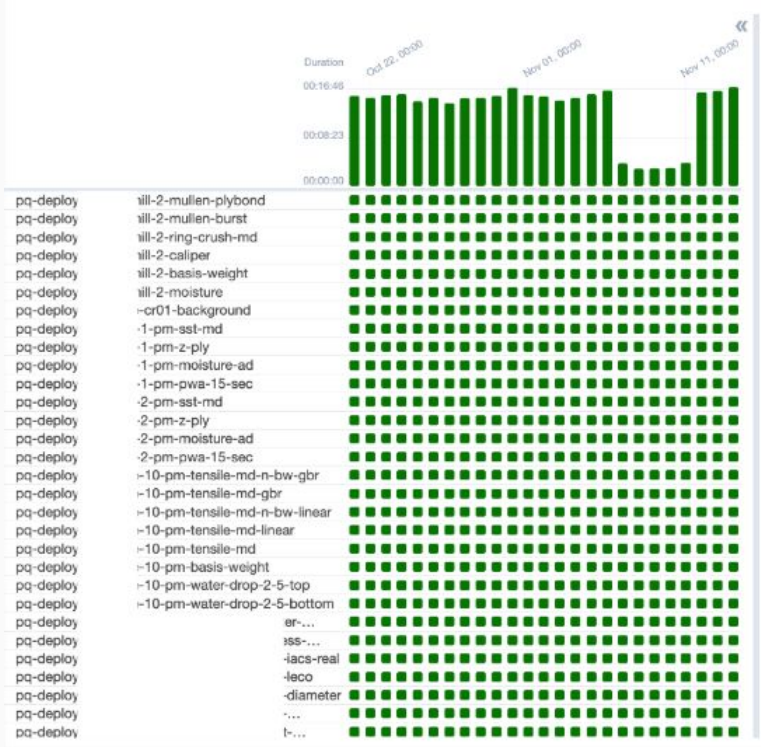
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MLFlow Artifacts



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Model Deployment via Apache Airflow



DAG
predictive_quality_deploy_v1

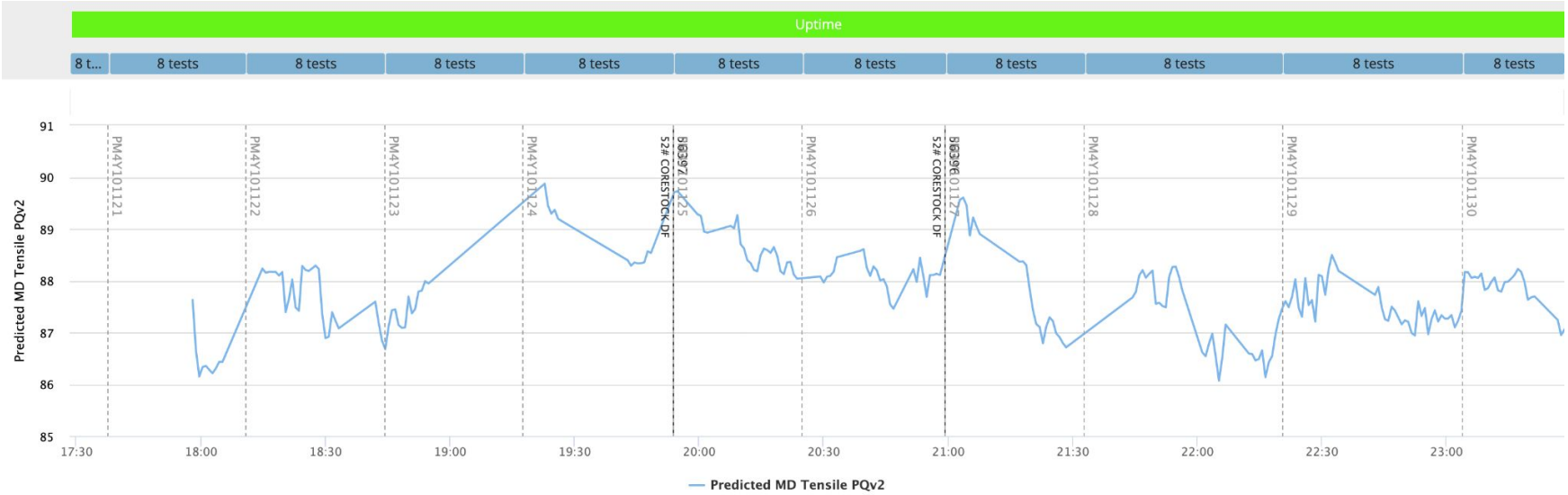
Details | Graph | Gantt | Code

DAG Runs Summary	
Total Runs Displayed	25
Total success	25
First Run Start	2024-10-22, 00:00:01 UTC
Last Run Start	2024-11-15, 00:00:01 UTC
Max Run Duration	00:16:46
Mean Run Duration	00:12:58
Min Run Duration	00:02:54
DAG Summary	
Total Tasks	37
MLFactoryCloudRunJobOperators	35
OdenGKEPodOperator	1
PassiveCheckOperator	1
DAG Details	
Dag id	predictive_quality_deploy_v1
Description	null

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Real-Time Predictions with Dataflow (Google Cloud)

- DataFlow job manages real-time inference/ posting predictions to the Oden Platform



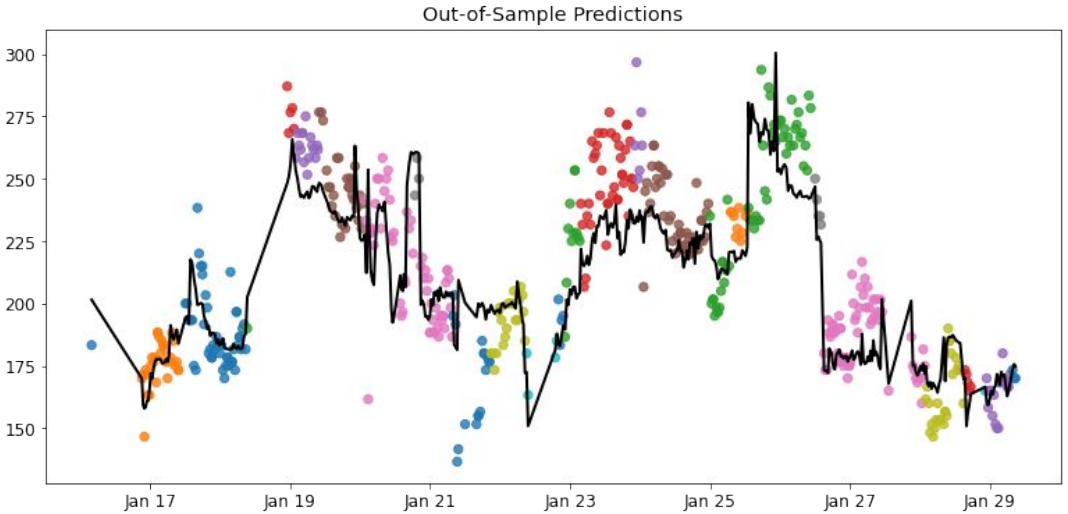
Key Variables Example: Paper

Predictive Quality Model Development

>90% accuracy

>70% correlation

<3 sigma error



Ply Bond - Paper	
Key Variable	**Coefficient**
#7 Vat Flow	**-6.5224**
#10 Vat Flow	**-4.1779**
Shared Waste Water	**-4.0135**
2nd Press Bottom Felt Vacuum	**-3.5552**
Bone Dry Weight (scanner)	**-3.3621**
Production Rate (tons/hr)	**-2.8738**
D Refiner HPDT	**2.2074**
1st Dryer Section Speed	**2.988**
C Refiner KW	**3.1313**
C Refiner HPDT	**3.1394**
% Target	**3.5729**
#1 Vat Lvl Output	**3.5746**
D Refiner KW	**4.0107**
Total HPDT	**8.8416**

Total Process Variables: **127**

Model Variables: **38**

Key Variables: **14**

Tangible Results for Pulp & Paper

Fortune 500 Paper Customer

Goal: Optimize speed & reduce input Costs while maintaining quality
Solution: Oden Process AI + Data Engine, first 6 months

3% Increase to Line Speed
($440K benefit/year)

5% Reduction in Material & Energy Costs
($505K benefit/year)

$900K+
Savings / Year



Tangible Results for Wire & Cable

Customer: #1 Electrical cable producer in North America

Goal: Increase production through higher line speeds

Solution: Oden Process AI

When: 90 days from go-live

Scope: 2 extrusion lines, many products

Who: 20 Operators of varying experience

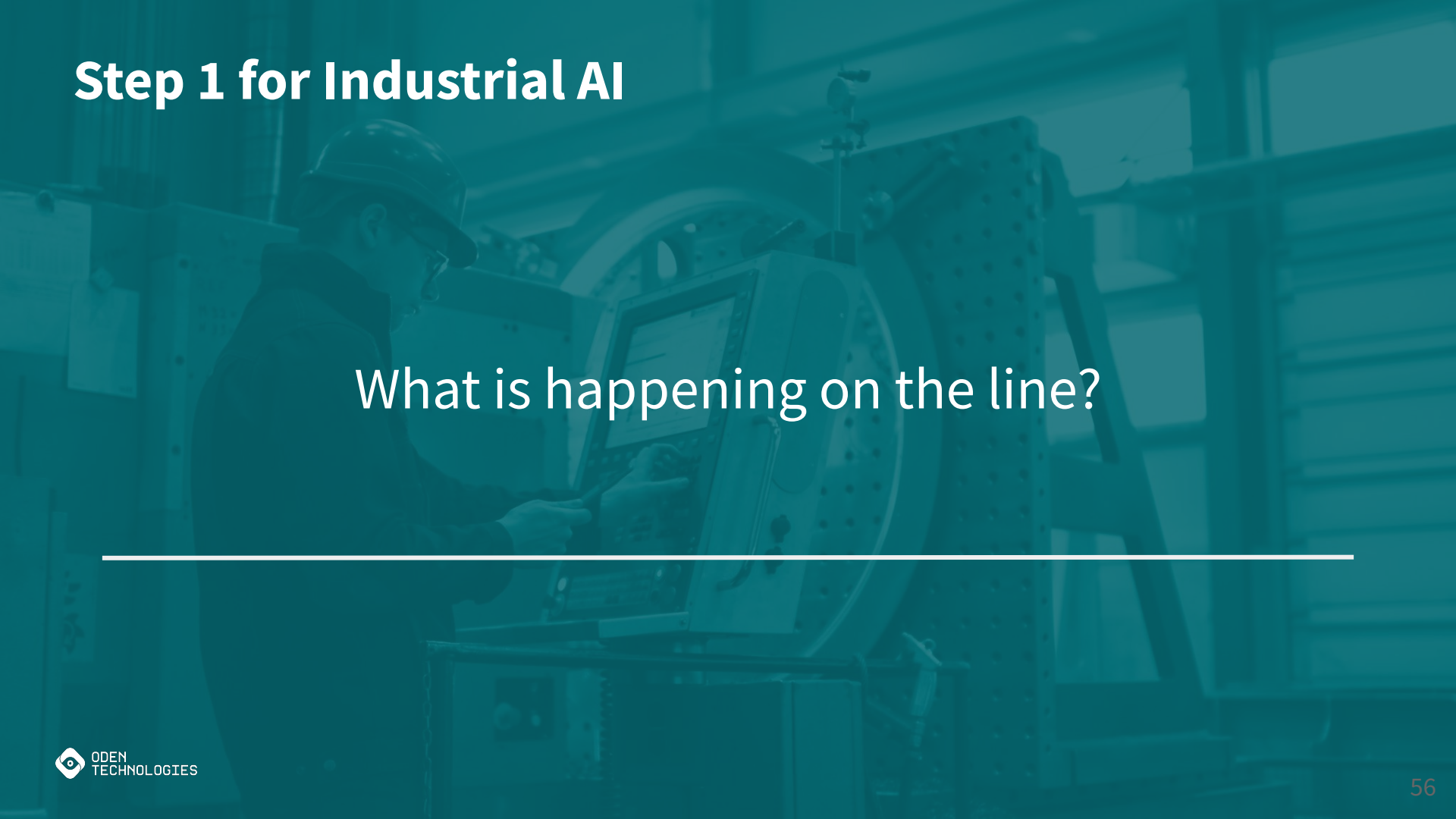
Results:

+20%
Increase in Production Rate

+5.3Mft
Additional Feet Produced over 90 days



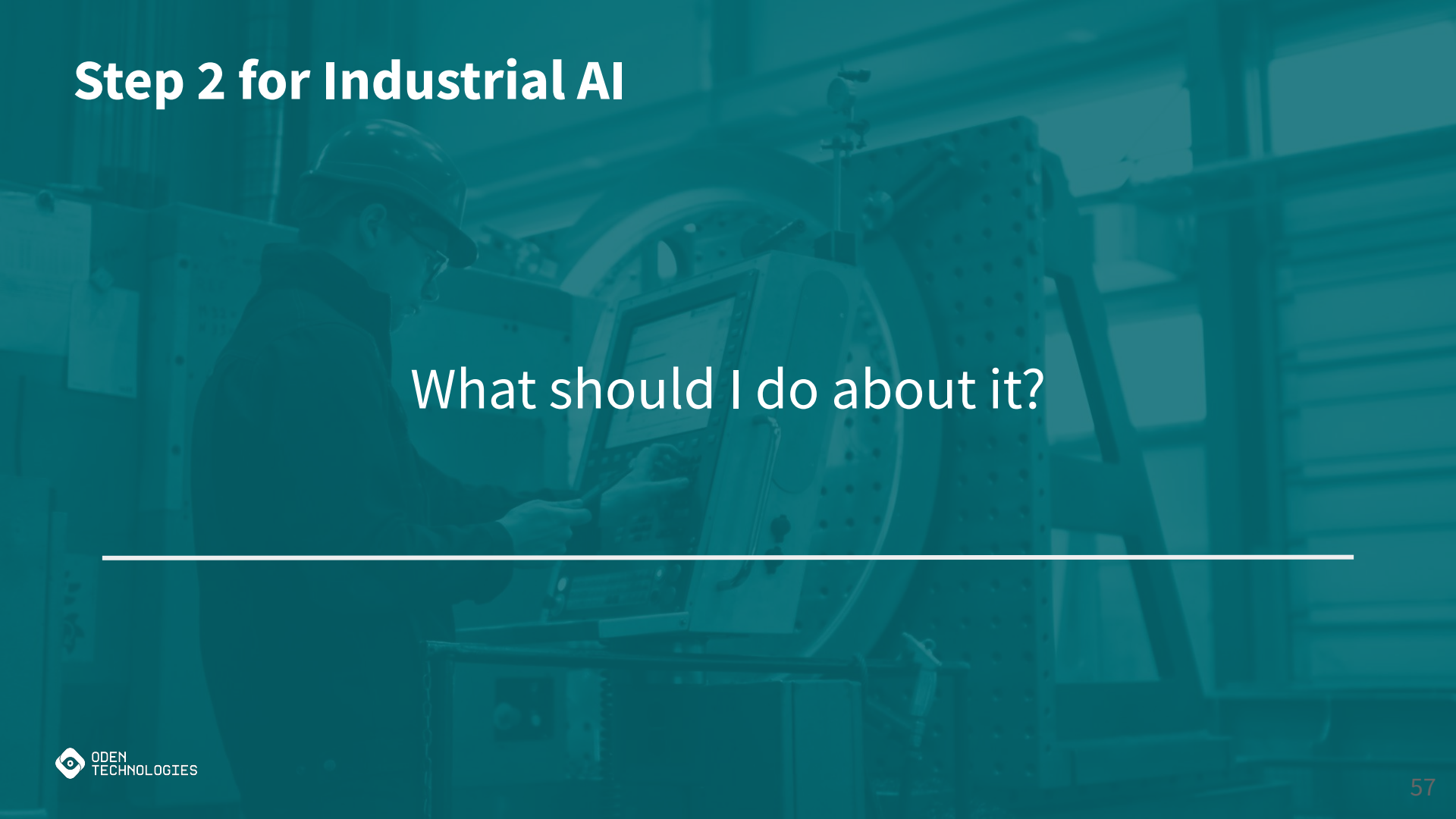
Step 1 for Industrial AI



What is happening on the line?

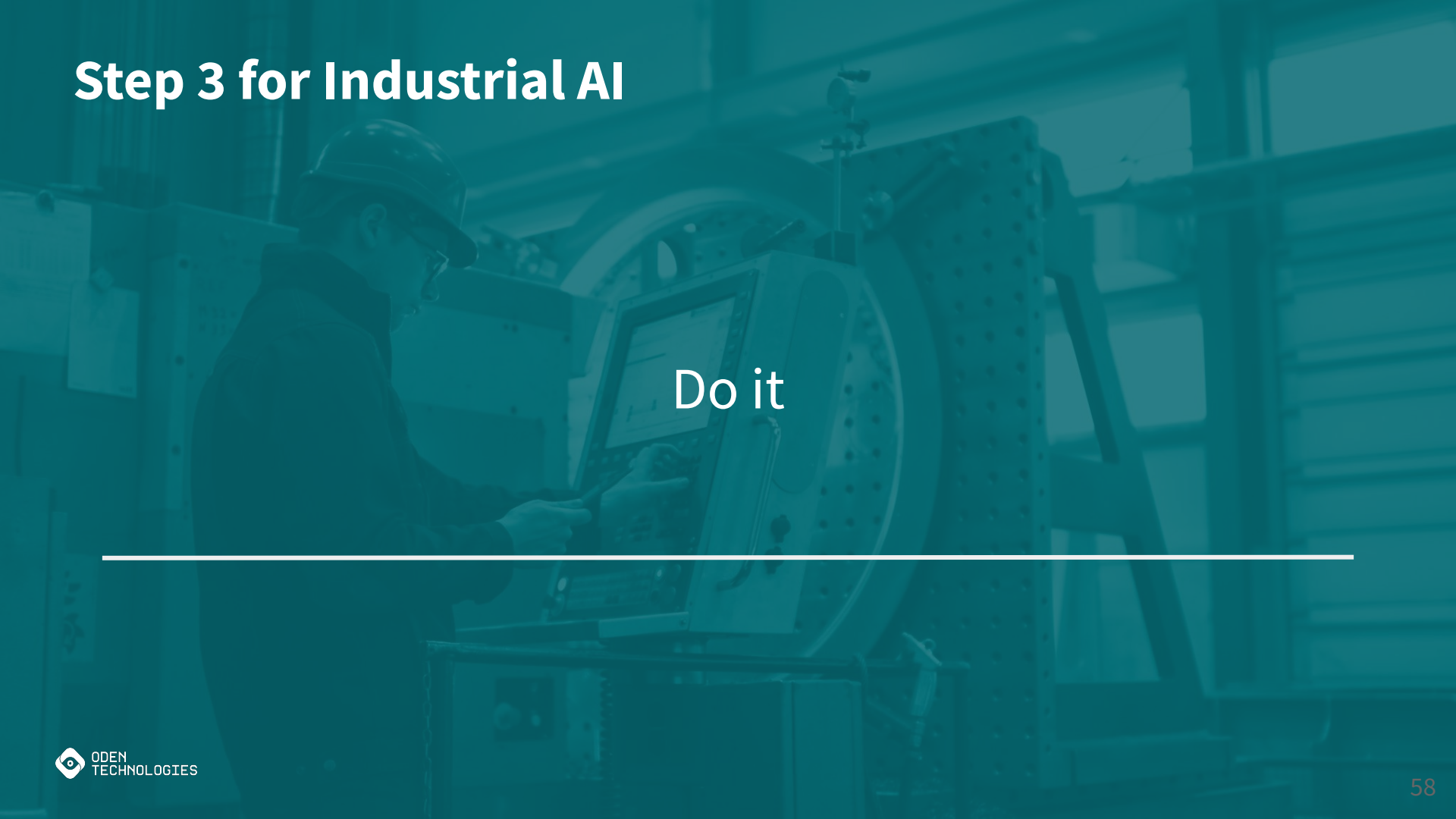
ODEN TECHNOLOGIES

Step 2 for Industrial AI



What should I do about it?

Step 3 for Industrial AI



Do it

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